End of year internship report

Analysis and Forecasting of Morocco's Outgoing International Mail Flows

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Dedication

I dedicate this work to my parents for their continuous support, encouragement, and sacrifices throughout my studies.

To my family and friends, for their patience, motivation, and confidence in me during the completion of this internship.

To all my teachers, who contributed to my education and helped me acquire the knowledge and skills necessary to reach this stage of my academic journey.

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Abstract

Barid Al-Maghrib manages large volumes of international postal shipments through operational information systems primarily designed for transaction processing and shipment tracking. While these systems provide detailed operational visibility, they offer limited support for predictive analytics and demand forecasting.

This internship addresses the problem of forecasting outgoing international mail flows in order to improve short-term operational planning. The work involved reconstructing historical time series from heterogeneous operational sources, developing a robust data preparation pipeline, and evaluating multiple forecasting approaches.

The proposed methodology follows an adaptation of the CRISP-DM framework and encompasses data extraction, preprocessing, feature engineering, model development, evaluation, and operationalization stages. Candidate forecasting models include statistical methods, machine learning algorithms, and deep learning architectures. Model performance was assessed using a rolling-origin cross-validation strategy and several complementary metrics.

Experimental results indicate that neural approaches, particularly BiTCN, achieve the best forecasting performances for the four-week planning horizon considered in this study. However, statistical models exhibit greater stability for longer forecasting horizons. To ensure coherence across aggregation levels, forecasts were reconciled using hierarchical forecasting techniques.

Finally, an interactive visualization system based on Altair was developed to enable users to explore historical observations and reconciled forecasts dynamically. The resulting framework constitutes a complete forecasting workflow capable of transforming operational postal data into actionable decision-support information and contributes to ongoing digital transformation initiatives within Barid Al-Maghrib.

Keywords: time series forecasting, hierarchical forecasting, postal logistics, CRISP-DM, BiTCN, NHITS, data engineering, Altair, Barid Al-Maghrib.

Abstract (FR)

Barid Al-Maghrib gère d'importants volumes d'envois postaux internationaux à travers des systèmes d'information principalement conçus pour le traitement transactionnel et le suivi des expéditions. Bien que ces systèmes offrent une visibilité opérationnelle détaillée, ils présentent des capacités limitées en matière d'analyse prédictive et de prévision de la demande.

Ce stage s'intéresse à la problématique de la prévision des flux de courrier international sortant afin d'améliorer la planification opérationnelle à court terme. Les travaux réalisés ont consisté à reconstruire des séries temporelles historiques à partir de sources opérationnelles hétérogènes, à développer une chaîne robuste de préparation des données et à évaluer plusieurs approches de prévision.

La méthodologie proposée s'appuie sur une adaptation de la démarche CRISP-DM et couvre les étapes d'extraction des données, de prétraitement, d'ingénierie des variables, de développement des modèles, d'évaluation et d'opérationnalisation. Les modèles étudiés comprennent des approches statistiques, des algorithmes d'apprentissage automatique et des architectures d'apprentissage profond. Les performances ont été évaluées à l'aide d'une validation croisée glissante et de plusieurs métriques complémentaires.

Les résultats expérimentaux montrent que les approches neuronales, notamment BiTCN, obtiennent les meilleures performances pour l'horizon de prévision de quatre semaines retenu dans cette étude. Toutefois, les modèles statistiques présentent une meilleure stabilité pour des horizons de prévision plus étendus. Afin de garantir la cohérence des prévisions entre les différents niveaux d'agrégation, une étape de réconciliation hiérarchique a été mise en œuvre.

Enfin, un système de visualisation interactif basé sur Altair a été développé afin de permettre l'exploration dynamique des observations historiques et des prévisions réconciliées. La solution proposée constitue ainsi une chaîne complète de prévision capable de transformer des données postales opérationnelles en informations exploitables pour l'aide à la décision, tout en s'inscrivant dans les initiatives de transformation numérique engagées par Barid Al-Maghrib.

Mots-clés : prévision de séries temporelles, prévision hiérarchique, logistique postale, CRISP-DM, BiTCN, NHITS, ingénierie des données, Altair, Barid Al-Maghrib.

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List of Abbreviations

Abbreviation	Definition
ABB	Al Barid Bank
API	Application Programming Interface
ARIMA	AutoRegressive Integrated Moving Average
BAM	Barid Al-Maghrib
BC	Barid Cash
BiTCN	Bidirectional Temporal Convolutional Network
CRISP-DM	Cross-Industry Standard Process for Data Mining
CRBT	Contre-Remboursement
CEC	Client en Compte
CCP	Compte Courant Postal
CCABB	Compte Courant Al Barid Bank
CES	Complex Exponential Smoothing
ETL	Extract, Transform, Load
ETS	Error, Trend, Seasonal
MEI	Mise en Instance
MED	Mise en Distribution
MFLES	Multiple Frequency Linear Exponential Smoothing
MASE	Mean Absolute Scaled Error
MinT	Minimum Trace Reconciliation
NHITS	Neural Hierarchical Interpolation for Time Series Forecasting
ND	Normalized Deviation
OLTP	Online Transaction Processing
PoD	Pay on Delivery
RPA	Robotic Process Automation
RMAE	Relative Mean Absolute Error
SMI	Système de Messagerie Intégré
SPIS	Scaled Pinball Interval Score
TBATS	Trigonometric, Box-Cox transformation, ARMA errors, Trend, and Seasonal components
UPU	Universal Postal Union

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General Introduction

Barid Al-Maghrib (BAM) is the national postal operator of Morocco and plays a central role in the provision of postal, logistics, financial, and digital services throughout the country. Through its extensive network and international partnerships, the organization manages large volumes of mail and parcel flows whose efficient handling is essential to maintaining service quality and operational performance.

Like many historical postal operators, BAM is currently engaged in a digital transformation process aimed at improving operational efficiency, increasing shipment traceability, and progressively adopting data-driven decision-making practices. Despite the availability of significant amounts of operational data, existing information systems remain primarily designed for transaction processing, shipment tracking, and operational reporting. Consequently, forecasting capabilities are not natively integrated into these systems, limiting the ability to anticipate future demand and support proactive planning activities.

This internship was carried out within BAM as part of the requirements for obtaining an engineering degree. A previous internship experience within the organization provided an initial exposure to its operational environment and information systems. This experience, combined with an interest in data engineering and predictive analytics, motivated the decision to pursue a second internship within the same institution and to address a more technically challenging problem involving large-scale data extraction and forecasting.

The central problem addressed in this internship is the absence of a forecasting framework capable of exploiting historical operational data to anticipate outgoing international mail volumes over short planning horizons. Existing planning activities rely mainly on descriptive reports and historical observations, making it difficult to proactively allocate resources, anticipate workload fluctuations, and optimize logistics operations.

The objective of this work is therefore to design and implement an end-to-end forecasting pipeline adapted to the constraints of BAM's information systems. More specifically, the study aims to:

- reconstruct historical shipment series from operational systems;
- develop a reproducible data preparation workflow;
- investigate statistical, machine learning, and deep learning forecasting approaches;
- evaluate forecasting performance using rolling-origin cross-validation;
- generate coherent forecasts across multiple aggregation levels through hierarchical reconciliation;
- provide an interactive visualization interface enabling users to explore and compare forecasting results.

More broadly, this internship seeks to demonstrate how modern data engineering practices and time-series forecasting techniques can be leveraged to bridge the gap between legacy operational information systems and data-driven decision-support tools.

The remainder of this report is organized as follows. Chapter 1 introduces Barid Al-Maghrib, its international mail activities, information systems, and the internship context. Chapter 2 presents the methodology adopted in this study following an adaptation of the CRISP-DM framework to the postal forecasting problem, covering data preparation, model development, evaluation procedures, and operationalization aspects. Chapter 3 describes the implementation of the proposed solution, discusses the experimental results, and presents the interactive visualization system developed to facilitate forecast exploration and support operational decision-making.

Chapter 1
Host Institution and Internship Context

Introduction

This chapter presents the institutional, operational, and technical environment in which the internship was conducted. Understanding the activities of Barid Al-Maghrib (BAM), particularly those related to international postal exchanges and the information systems supporting them, is necessary to situate the study within its professional context and motivate the forecasting problem addressed in this work.

The chapter first introduces Barid Al-Maghrib, its missions, organizational structure, and role within the Moroccan postal sector. It then describes the characteristics of international mail activities and the information systems used to support postal operations and monitor shipment flows. Particular attention is given to the operational constraints encountered when exploiting historical data originating from these systems.

The internship context is subsequently presented, highlighting the objectives assigned during the placement, the technologies employed, and the analytical workflow progressively developed throughout the study. Finally, the problem addressed during the internship is formulated, together with the objectives pursued, which provide the basis for the methodological developments discussed in the following chapters.

1.1 Presentation of Barid Al-Maghrib

Barid Al-Maghrib (BAM) is Morocco's designated postal operator and is responsible for ensuring the provision of universal postal services throughout the national territory. In addition to its traditional postal activities, the institution has progressively diversified its operations to include logistics, financial, and digital services. BAM currently operates as a state-owned company while remaining entrusted with public service missions defined by Moroccan postal legislation [2], [3].

1.1.1 Historical Background

The origins of postal services in Morocco date back to the late nineteenth century. A structured postal organization was established in 1892, laying the foundations for a national communication network [4].

During the Protectorate period, postal, telegraph, and telephone services were reorganized and integrated into the Administration Chérifienne des Postes, des Télégraphes et des Téléphones, created in 1912. Following Moroccan independence in 1956, these services were placed under ministerial supervision, and efforts were undertaken to extend postal coverage throughout the country.

In 1984, the Office National des Postes et des Télécommunications (ONPT) was established to consolidate postal and telecommunications activities within a single public institution [4]. The communications sector was later reformed through Law 24-96, which separated postal and telecommunications activities and created Barid Al-Maghrib as an autonomous public institution endowed with legal personality and financial independence [2].

A subsequent reform transformed BAM into a state-owned limited company in 2010, reinforcing its managerial autonomy while preserving public ownership. This reform was accompanied by the consolidation of financial activities under Al Barid Bank (ABB) [3].

1.1.2 Missions of Barid Al-Maghrib

As the national postal operator, BAM is responsible for ensuring universal postal service coverage and maintaining the infrastructure necessary for the collection, sorting, transportation, and distribution of postal items within Morocco and abroad. Its activities encompass ordinary and registered mail, express services, parcel logistics, financial services, and digital solutions intended to support citizens, businesses, and public administrations [2].

BAM also represents Morocco within international postal organizations and ensures compliance with the technical and operational standards governing international postal exchanges.

1.1.3 Postal Activities

Although traditional letter mail remains part of BAM's historical mission, parcel and logistics services have become increasingly important due to the rapid growth of e-commerce and cross-border trade.

Among these activities, international shipments constitute a strategic segment because they involve coordination between postal agencies, sorting centers, transportation providers, customs administrations, and foreign postal operators. Efficient management of these flows requires accurate monitoring tools as well as mechanisms capable of anticipating variations in shipment volumes.

Since the objective of this internship was to analyze and forecast outgoing international mail volumes, particular attention was devoted to this activity, which is discussed in the following section.

1.1.4 Subsidiaries and Group Structure

BAM has progressively adopted a group structure aimed at diversifying its activities and adapting to changes in the postal and financial sectors.

Its principal subsidiary is Al Barid Bank (ABB), established in 2010 to consolidate postal financial services and promote financial inclusion. Leveraging BAM's nationwide network, ABB provides banking services to individuals and small businesses throughout Morocco [5].

Other affiliated entities contribute to logistics, express delivery, and media-related activities, enabling the group to diversify its sources of revenue and strengthen its market position.

1.1.5 International Cooperation

Barid Al-Maghrib is a member of the Universal Postal Union (UPU), the specialized agency of the United Nations responsible for coordinating international postal exchanges and defining common operational standards [6].

Through this membership, Morocco participates in a global postal network connecting more than 180 countries. The standards established by the UPU govern shipment preparation, customs procedures, electronic data exchange, tracking interoperability, and compensation mechanisms between designated postal operators [7].

Compliance with these standards ensures the interoperability of Moroccan postal services with foreign networks and contributes to improving the reliability of international mail operations.

1.1.6 Current Challenges

Like many postal operators worldwide, BAM faces declining volumes of traditional correspondence while simultaneously managing growing parcel traffic and increasing customer expectations regarding shipment traceability.

At the same time, operational information systems have historically been designed to support day-to-day processing activities rather than analytical applications. Consequently, despite the availability of large volumes of historical data, anticipating future shipment volumes remains challenging.

This context motivated the work carried out during the internship, which focused on transforming operational postal data into datasets suitable for analysis and evaluating forecasting methods capable of supporting planning activities.

1.2 International Mail Activities

International mail activities constitute a key component of Barid Al-Maghrib's operations. They enable the exchange of letters, documents, and parcels between Morocco and foreign destinations through a network governed by international agreements and operational standards defined mainly by the Universal Postal Union (UPU) [6], [8].

The increase in e-commerce activity, the intensification of international trade, and the role of the Moroccan diaspora have significantly increased both the volume and variability of outgoing international shipments. This evolution has made the management of international flows more complex, requiring improved monitoring, planning, and forecasting capabilities.

1.2.1 International Postal Framework

International postal exchanges are organized under the Universal Postal Union (UPU), which defines standardized procedures ensuring interoperability between national postal operators. These standards regulate the preparation, routing, tracking, and settlement of international mail flows [7].

Within this framework, Barid Al-Maghrib is responsible for ensuring the proper handling of outgoing international shipments, including compliance with customs regulations, operational standards, and electronic data exchange requirements.

1.2.2 Structure of International Flows

International postal flows are divided into two main categories:

Outgoing international mail: shipments sent from Morocco to foreign destinations; Incoming international mail: shipments received from abroad.

The present internship focuses exclusively on outgoing international mail, which constitutes the primary dataset used for forecasting.

The routing of shipments depends on several operational factors, including destination country, transport availability, service level, and customs constraints.

1.2.3 International Postal Products

Outgoing international mail is composed of several service categories with different levels of tracking, delivery speed, and handling priority.

Ordinary international mail includes letters and low-value parcels with limited tracking capabilities. Registered mail provides identification and tracking at key stages of the shipment lifecycle, making it suitable for administrative documents and sensitive items.

The Express Mail Service (EMS) is a premium service offering priority handling, reduced transit times, and end-to-end tracking, coordinated at the international level within the UPU framework [9]. International parcels, strongly driven by e-commerce growth, represent the most dynamic segment due to increasing cross-border consumer activity.

1.2.4 Operational Flow of Outgoing Mail

Outgoing international shipments follow a sequence of operational stages before reaching destination countries.

Shipments are first accepted at post offices or authorized collection points, where key information such as sender details, destination country, weight, and service type is recorded. Each item is assigned a unique identifier used throughout its lifecycle.

Items are then routed to sorting facilities, where they are grouped by destination and prepared for dispatch. Depending on the nature of the shipment, customs procedures may be applied, particularly for goods requiring declaration or inspection.

After sorting and consolidation, shipments are transported—primarily by air—to destination countries or transit hubs. Once they arrive abroad, responsibility is transferred to the destination postal operator, which completes the final delivery process.

1.2.5 Operational Characteristics

Outgoing international mail flows are characterized by strong geographical concentration, with a significant share of shipments originating from major urban centers such as Casablanca, Rabat, Marrakech, and Tangier. Destination countries are diverse, although European countries represent a substantial proportion of total flows due to historical, economic, and migratory links [10].

The analysis conducted during this internship also highlighted clear seasonal patterns, with recurring peaks observed during specific periods such as the weeks preceding Ramadan. These variations reflect both cultural factors and changes in international commercial activity.

Other external drivers include e-commerce promotions, holiday periods, transportation disruptions, and variations in international trade conditions.

1.2.6 Operational Challenges

Managing international mail flows requires addressing several operational challenges. Demand variability complicates transportation planning and resource allocation, particularly during peak periods. In addition, customs procedures introduce uncertainty due to heterogeneous processing times across destination countries.

From a data perspective, operational systems primarily support tracking and execution rather than analytical processing. As a result, historical data must be reconstructed and reorganized before it can be used for forecasting purposes.

This gap between operational systems and analytical requirements motivates the methodological approach adopted in this internship, which focuses on transforming raw operational data into structured time series suitable for forecasting.

1.3 Information Systems

The information systems used by Barid Al-Maghrib form the main operational backbone for managing postal activities and constitute the primary source of data used in this internship. They support shipment processing, operational monitoring, and the consultation of activity reports used by internal teams for daily decision-making.

Although these systems contain rich historical information, they are primarily designed for operational execution rather than analytical exploitation. This design choice has a direct impact on how data can be accessed, extracted, and reused for forecasting purposes.

1.3.1 Digital Transformation Context

Like many postal operators, Barid Al-Maghrib is undergoing gradual digital transformation to handle increasing parcel volumes, improve traceability, and support operational monitoring. The growth of e-commerce and the rising expectations of users regarding shipment tracking have reinforced the importance of reliable and structured data.

However, a large part of the existing information systems still follows an operational logic, where data is optimized for transaction processing rather than large-scale analysis. This creates a gap between available information and its direct usability for statistical modeling.

Within this context, the internship required the development of dedicated data preparation and extraction procedures in order to transform operational records into analytical datasets.

1.3.2 Système de Messagerie Intégré (SMI)

The Système de Messagerie Intégré (SMI) is the central application used for managing postal shipments and recording operational events throughout their lifecycle.

It allows users to:

- Register shipment information at acceptance;
- Track processing stages across postal facilities;
- Consult shipment histories;
- Monitor sorting and dispatch operations;
- Access operational reporting interfaces.

While SMI contains the core historical data used in this study, it presents several limitations from an analytical perspective. In particular, it does not provide a dedicated API for bulk extraction, and its export functionalities are limited in scope and performance. Long queries covering extended time periods are also constrained by system responsiveness.

These limitations had a direct influence on the data acquisition strategy used during the internship.

1.3.3 Tracking and Event Systems

Shipment tracking is based on event logs associated with each parcel or mail item. Each event corresponds to a timestamped operational action such as acceptance, sorting, dispatch, transit, or delivery.

This event-based structure makes it possible to reconstruct the lifecycle of each shipment. However, accessing these events at scale requires navigating interfaces that are primarily designed for individual shipment queries rather than batch extraction.

As a result, extracting historical tracking data required automated interaction with system interfaces in order to collect large volumes of shipment-level information.

1.3.4 Reporting Interfaces

In addition to operational systems, Barid Al-Maghrib provides reporting tools that display aggregated indicators related to postal activity. These include daily volumes, destination distributions, and agency-level statistics.

These reporting interfaces are useful for operational monitoring but remain limited for analytical work. In particular, they do not systematically provide raw historical data in formats directly suitable for modeling or time-series construction.

This constraint made it necessary to rely on automated extraction pipelines to retrieve and structure the required information.

1.3.5 Data Extraction Approach

Given the absence of direct programmatic access to historical data, an automated extraction process was implemented using browser automation techniques with Playwright for Python [11].

This approach allowed:

- Navigation of reporting and tracking interfaces;
- Automated selection of time intervals and filters;
- Extraction of shipment-level and tracking data;
- Iterative collection over long historical periods.

The extracted data was then consolidated and stored in a structured format suitable for downstream processing and analysis.

1.3.6 Data Engineering Constraints

Several technical limitations were encountered during data collection:

- No public API for historical data access;
- Interface-based access requiring manual or automated interaction;
- Performance limitations over large date ranges;
- Heterogeneous export formats depending on the module used.

To address these constraints, the data pipeline included automated retries, incremental extraction over smaller time windows, and intermediate storage to ensure robustness.

Final datasets were stored in columnar format (Parquet) to improve efficiency during analysis and model training.

1.3.7 Role of Information Systems in This Internship

The information systems described above constitute both the origin and the main constraint of the analytical work performed in this internship.

They provide detailed operational data on shipment processing, but require significant transformation before being usable for forecasting. This transformation step represents a central part of the methodology developed in this work.

The quality, structure, and accessibility of these systems directly influenced the design of the datasets used in the subsequent modeling phase.

1.4 Internship Context

This internship was carried out within Barid Al-Maghrib as part of an engineering degree program. The work focused on the analysis and forecasting of outgoing international mail volumes using historical operational data extracted from internal information systems.

The project sits at the intersection of data engineering and applied forecasting. It required both the construction of usable datasets from operational systems and the development of predictive models adapted to the structure and limitations of the available data.

1.4.1 Industrial Context

International mail represents a structurally important activity for Barid Al-Maghrib due to the continuous growth of cross-border exchanges. This growth is mainly driven by e-commerce and the volume of shipments linked to Moroccan communities abroad.

In this context, anticipating shipment volumes becomes relevant for operational planning. Variations in demand directly affect transport organization, sorting capacity, and workforce allocation.

The forecasting problem addressed in this internship is therefore not purely analytical; it is tied to practical operational needs related to planning and resource management.

1.4.2 Working Environment

The internship was conducted in an environment strongly centered on operational systems. These systems are designed primarily to support shipment processing and tracking rather than analytical exploration.

Several characteristics of this environment shaped the work:

- Data is distributed across multiple interfaces rather than centralized;
- Historical access is limited and often constrained by performance issues;
- Extraction functionalities are designed for manual consultation;
- Reporting tools are oriented toward operational monitoring rather than analysis.

As a result, a significant portion of the internship was dedicated to making the data usable for analytical purposes before any modeling could be performed.

1.4.3 Internship Mission

The main objectives of the internship can be summarized as follows:

- Reconstruct historical datasets from operational systems;
- Develop automated procedures for data extraction;
- Clean and structure raw shipment and tracking data;
- Analyze temporal patterns in outgoing international mail flows;
- Build and evaluate forecasting models;
- Compare statistical, machine learning, and deep learning approaches.

These objectives were pursued in an iterative manner, where data preparation and modeling informed each other.

1.4.4 Tools and Technologies

The internship relied on a set of tools chosen to handle both data extraction and time-series analysis tasks:

- Python for data processing and model implementation;
- Playwright for automated extraction from web-based systems [11];
- Data manipulation libraries for cleaning and transformation tasks;
- Columnar storage formats (Parquet) for efficient handling of large datasets;
- Standard time-series forecasting frameworks for model evaluation.

No dedicated dashboarding or business intelligence layer was used in the scope of this work, as the focus remained on dataset construction and modeling rather than deployment.

1.4.5 Data Context

The datasets used in this internship originate mainly from the Système de Messagerie Intégré (SMI) and related operational reporting interfaces.

They include information such as:

- Shipment acceptance dates;
- Destination countries;
- Processing and tracking events;
- Operational status updates;
- Additional attributes used for exploratory analysis.

Access to these data sources was constrained by several limitations:

- Lack of direct programmatic access;
- Limited export capabilities;
- Performance issues over long time intervals;
- Inconsistent historical coverage across modules.

To address these constraints, an extraction pipeline was developed to progressively collect, consolidate, and structure the data into analyzable formats.

1.4.6 Position of the Internship

Within Barid Al-Maghrib, this work is positioned as an exploratory analytical effort aimed at reusing operational data for forecasting purposes.

Rather than modifying existing systems, the internship focuses on extracting value from existing infrastructure by transforming operational records into structured time series.

This approach highlights the potential of internal data for supporting planning activities, even when systems were not originally designed for analytical use.

1.4.7 Contribution

The main contribution of the internship lies in the development of a complete workflow that connects operational systems to forecasting models.

This workflow includes:

- Automated extraction of shipment data from operational interfaces;
- Transformation and cleaning of raw records;
- Construction of consistent time series;
- Application of multiple forecasting approaches;
- Evaluation of predictive performance across models.

The resulting pipeline demonstrates how operational postal data can be repurposed for analytical and predictive tasks, despite the constraints of legacy information systems.

1.4.8 Summary

This internship is situated in a context where operational information systems impose significant constraints on data access and analysis.

Despite these limitations, it was possible to construct structured datasets and apply forecasting techniques to outgoing international mail volumes, providing a basis for better understanding and anticipating shipment dynamics.

1.5 Problem Statement and Objectives

The work carried out during this internship addresses the problem of forecasting outgoing international mail volumes using historical operational data from Barid Al-Maghrib.

Although operational information systems provide access to shipment tracking and reporting functionalities, they are primarily designed for transaction processing and operational monitoring. As a result, they offer limited support for historical analysis and almost no direct capabilities for predictive modeling. This creates a gap between available data and its analytical usability.

Bridging this gap requires transforming operational records into structured datasets suitable for time-series analysis. This involves extracting data from heterogeneous systems, cleaning and consolidating it, and reconstructing consistent temporal series that can be used for forecasting models.

1.5.1 Operational Need for Forecasting

Outgoing international mail volumes are characterized by strong variability over time. Exploratory analysis conducted during this internship revealed recurring seasonal patterns, particularly increases in activity during periods preceding Ramadan. These variations reflect both cultural and commercial dynamics affecting postal demand.

Several external factors also contribute to fluctuations in shipment volumes:

- Growth of e-commerce and cross-border purchases;
- Variations in international transport capacity;
- Customs processing delays in destination countries;
- Seasonal migration and exchanges with Moroccan communities abroad.

From an operational perspective, these variations directly affect workload planning, transport scheduling, and resource allocation. Being able to anticipate such changes is therefore valuable for improving operational efficiency.

1.5.2 Data-Related Constraints

A major difficulty encountered during the internship concerns data accessibility.

The main limitations include:

- Restricted export capabilities from operational systems;
- Absence of direct APIs for historical extraction;
- Dependence on interfaces designed for manual consultation;
- Performance degradation when querying long time periods;
- Inconsistent coverage across different system modules.

Because of these constraints, constructing a usable dataset required significant preprocessing effort. This included automated extraction procedures, consolidation of multiple data sources, and data cleaning operations to ensure consistency and reliability.

1.5.3 Analytical Gap

Operational systems provide detailed transactional visibility but do not directly support forecasting tasks. They answer the question of what happened, but not what is likely to happen next.

The analytical gap can be summarized as follows:

- Historical data exists but is fragmented across systems;
- Time series must be reconstructed from event-level records;
- Data quality issues must be addressed before modeling;
- Forecasting requires aggregation and temporal alignment that are not provided natively.

The objective of this work is not to replace operational systems, but to reuse their data for analytical purposes.

1.5.4 Problem Formulation

The forecasting problem can be formulated as a univariate time-series prediction task.

Let y_t denote the observed volume of outgoing international mail at time t, and let h represent the forecasting horizon. The goal is to estimate future values $y_{\{t+h\}}$ based on historical observations.

This can be expressed as:

$$\hat{y}_{t+h} = f(y_t, y_{t-1}, ..., y_{t-n})$$

where f is a forecasting function learned from past data.

The problem is made more challenging by several characteristics of the data:

- Seasonal patterns and periodic fluctuations;
- Irregularities and missing observations;
- Structural changes in shipment volumes over time;
- Increased uncertainty for longer forecasting horizons.

In this study, emphasis is placed on short-term forecasting, as it is most relevant for operational planning.

1.5.5 Objectives of the Study

The objectives of this internship are as follows:

- Construct a consistent historical dataset from operational sources;
- Analyze temporal patterns in outgoing international mail flows;
- Develop and compare forecasting models;
- Evaluate model performance using statistical metrics;
- Assess the operational relevance of forecasting outputs.

These objectives aim to demonstrate how operational postal data can be transformed into actionable analytical information.

1.5.6 Expected Outcome

The expected outcome is the development of a complete analytical workflow covering:

- Data extraction and consolidation;
- Data preprocessing and transformation;
- Forecast generation using multiple modeling approaches;
- Evaluation of predictive performance.

Beyond the numerical results, the goal is to establish a reproducible methodology for exploiting operational postal data in a forecasting context.

1.5.7 Summary

The problem addressed in this internship arises from the mismatch between operational data systems and analytical requirements.

By combining data engineering and time-series forecasting techniques, this work investigates the feasibility of predicting outgoing international mail volumes and their potential value for supporting operational decision-making within Barid Al-Maghrib.

Conclusion

This chapter presented the institutional and operational context in which the internship was carried out. It introduced Barid Al-Maghrib, its international mail activities, and the information systems used to manage and monitor postal operations.

The study of this environment made it possible to identify several challenges related to the exploitation of historical data, including limited extraction capabilities, the predominance of operational systems, and the difficulty of directly using available data for forecasting purposes. These observations justified the implementation of dedicated data collection and preprocessing procedures as a prerequisite for developing forecasting models.

The following chapter describes the methodology adopted for data acquisition, preparation, and exploratory analysis, as well as the forecasting approaches implemented and evaluated in the context of outgoing international mail volume prediction.

Chapter 2
Methodology

Introduction

This chapter presents the methodological framework adopted to address the forecasting problem defined in Chapter 1. The approach is based on the CRISP-DM process model [12], adapted to the constraints of Barid Al-Maghrib's operational information systems and to the specific requirements of hierarchical time-series forecasting.

In classical data science applications, CRISP-DM assumes the availability of structured and centralized datasets. In the present context, however, the data originates from operational postal systems originally designed for transaction processing rather than analytical use. This leads to fragmented data sources, heterogeneous interfaces, and limited direct access to structured historical datasets.

To address these constraints, the methodology implemented in this internship defines a complete pipeline covering data extraction, hierarchical dataset construction, feature engineering, model development across multiple forecasting paradigms, reconciliation of hierarchical outputs, and evaluation using time-series cross-validation. A final deployment-oriented component is also introduced through an interactive visualization layer built with Altair, enabling exploration and interpretation of forecasting results in an operational context.

The global structure of this pipeline is summarized in Figure 2.1.

2.1 CRISP-DM Framework Adaptation

The methodology adopted in this internship is structured according to the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework [12], which remains one of the most widely used reference models for data science and machine learning projects in industrial contexts.

CRISP-DM defines a cyclical process composed of six phases: Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, and Deployment. While this structure is general-purpose, it is particularly well suited for projects involving heterogeneous data sources and iterative model development, which is the case for hierarchical forecasting of postal flows at Barid Al-Maghrib.

However, in this internship, CRISP-DM is not applied in a strictly sequential manner. Instead, it is adapted into a more engineering-oriented pipeline that reflects the operational constraints of postal information systems and the iterative nature of model development. In particular, the separation between modeling and evaluation is iterative, as multiple forecasting paradigms are tested and compared continuously.

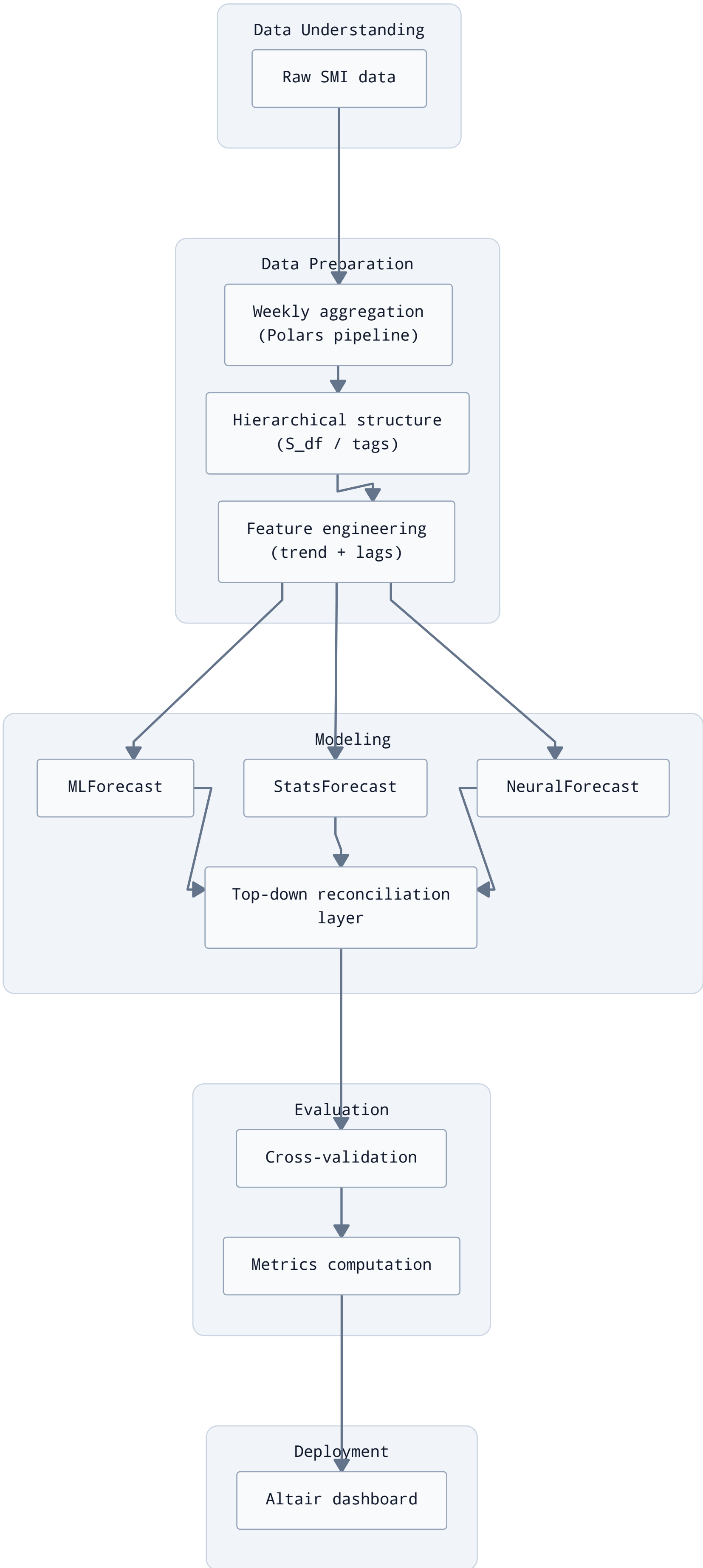

Figure 2.1 - Global architecture of the forecasting pipeline adapted from CRISP-DM.

Figure 2.1 illustrates the global architecture of the forecasting system implemented in this internship. The pipeline begins with raw operational data extracted from Barid Al-Maghrib's information systems and proceeds through a structured transformation process.

The first stage corresponds to data preparation, where weekly aggregation and hierarchical structuring are performed. This step is essential because the original data is not directly available in an analytical format and must be reconstructed into consistent time series.

The second stage introduces feature engineering, where both shared and model-specific features are generated. A key element is the trend component, which is consistently used across machine learning and neural forecasting models, ensuring comparability between different model families.

The third stage represents the modeling layer, where three distinct paradigms are applied: statistical models (StatsForecast), machine learning models (MLForecast), and deep learning models (NeuralForecast). This multi-paradigm design allows the comparison of fundamentally different forecasting approaches under a unified framework.

Finally, the pipeline includes a reconciliation step ensuring coherence across hierarchical levels, followed by evaluation and visualization. The presence of the visualization layer highlights that the system is not limited to offline modeling but also supports interactive analysis through an Altair-based interface, which is used for exploratory validation and operational interpretation of results.

2.2 Business Understanding

This section defines the operational context and objectives of the forecasting task addressed in this internship. In accordance with the CRISP-DM methodology [12], the Business Understanding phase aims to translate an industrial need into a formal data science problem that can be addressed using statistical and machine learning methods.

In the context of Barid Al-Maghrib, the main operational objective is the ability to anticipate fluctuations in outgoing international mail flows. These flows are directly influenced by a combination of structural and seasonal factors, including e-commerce activity, diaspora-related demand, and logistics constraints affecting international transport capacity.

Unlike standard prediction problems in controlled environments, postal demand forecasting is constrained by operational realities such as limited storage capacity in sorting centers, variability in international transport schedules, and heterogeneous behavior across destinations. These constraints require forecasts that are not only accurate, but also stable and interpretable from an operational perspective.

The operational scope of international mail forecasting is summarized in Figure 2.2, which presents the main factors influencing outgoing international mail demand and their impact on planning activities.

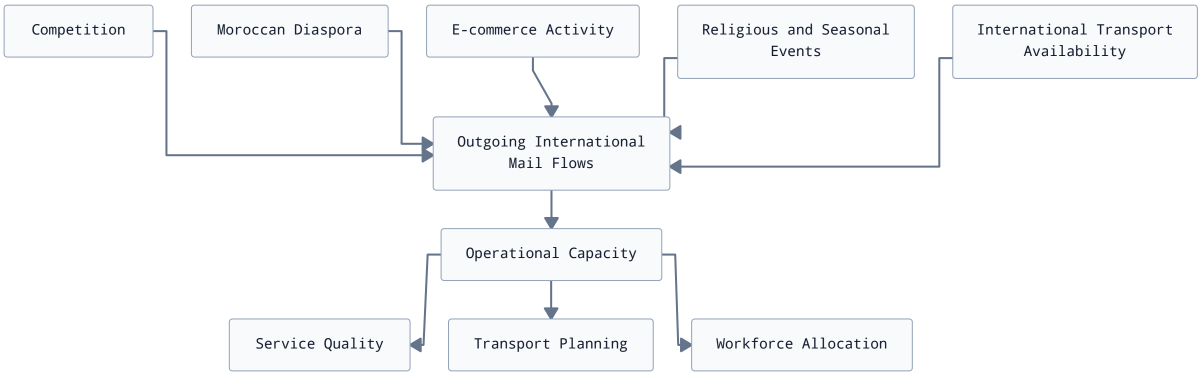

Figure 2.2 - Main factors influencing outgoing international mail flows and their operational implications.

Figure 2.2 shows that outgoing international mail volumes are affected by both internal and external factors. Demand-side drivers such as e-commerce activity, diaspora-related shipments, and seasonal events contribute to fluctuations in mail volumes, while supply-side constraints including transport availability and customs procedures influence the operator's ability to process these flows efficiently. Anticipating these variations is therefore essential for improving transport planning, workforce allocation, and maintaining service quality.

2.3 Exploratory Data Analysis

According to the CRISP-DM methodology [12], the Data Understanding phase aims to identify available data sources, assess their suitability for the forecasting task, and characterize the statistical properties of the target variable before any transformation or modeling step is undertaken.

In this internship, the forecasting task relies on operational data originating from Barid Al-Maghrib's information systems. More specifically, the study focuses on historical records associated with outgoing international shipments deposited in Morocco and processed through the Système de Messagerie Intégré (SMI).

The available data is transactional in nature, with each observation describing an individual shipment and containing operational information recorded during the acceptance process. Among the attributes used in this study are shipment identifiers, deposit dates, destination countries, originating agencies, and shipment weights.

As the objective is to forecast aggregate international mail flows rather than individual shipments, the raw observations are transformed into weekly time series through aggregation procedures described in the following section. The forecasting target considered in this work corresponds primarily to the total weight of outgoing shipments aggregated over one-week periods. Shipment counts are also retained as complementary indicators for exploratory analyses and consistency checks.

The historical data available for this internship covers shipments deposited from January 2023 onwards. Prior to any modeling effort, an exploratory analysis was conducted to understand the temporal behavior of international mail flows and to identify potential sources of variability.

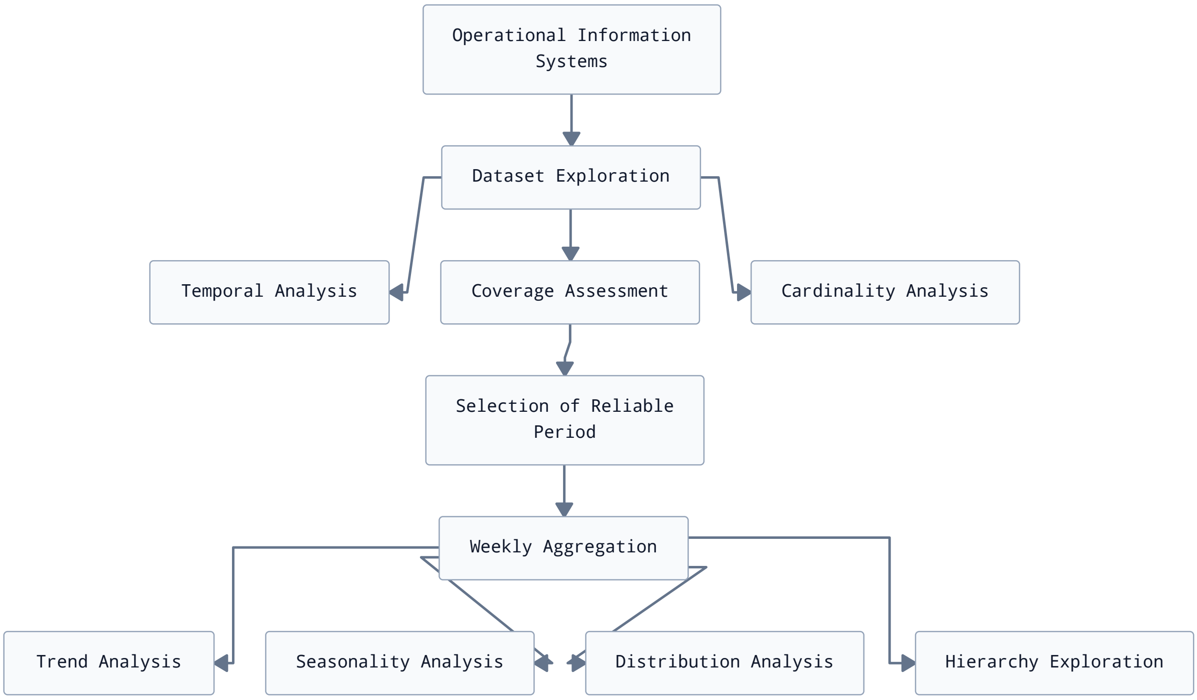

Figure 2.3 - Data understanding workflow implemented during the internship.

Figure 2.3 summarizes the activities carried out during the data understanding phase. Since the available information systems were primarily designed for operational purposes, an initial investigation was required to assess dataset coverage, data quality, and temporal consistency. This phase included the analysis of feature cardinalities, the identification of reliable observation periods, and the exploration of hierarchical structures before aggregating shipment records into weekly time series suitable for exploratory analyses and forecasting.

The exploratory analysis highlighted several characteristics of the dataset that influenced subsequent methodological choices.

First, the series exhibits a clear long-term trend associated with the increasing importance of parcel logistics and international exchanges. Second, recurrent seasonal patterns can be observed, particularly around religious and vacation periods. Preliminary analyses also suggested that the seasonality of international flows may be better represented using a Hijri annual cycle than a conventional Gregorian yearly seasonality. This observation motivated the use of a seasonal period of approximately fifty-one weeks in several forecasting models.

Finally, the exploratory analysis confirmed the existence of a natural hierarchical structure within the data. Shipments can be analyzed at multiple aggregation levels, including the national total, deposit centers, destination countries, or combinations of both dimensions. This characteristic makes hierarchical forecasting techniques particularly relevant for the present study.

2.4 Data Preparation

The operational data extracted from Barid Al-Maghrib's information systems cannot be directly used for forecasting purposes. The available records correspond to individual shipment transactions and are primarily intended to support operational processes such as acceptance, routing, and tracking. Consequently, several preprocessing steps were necessary to obtain coherent time series adapted to the forecasting task.

The preparation process implemented during this internship is summarized in Figure 2.4.

Figure 2.4 presents the preprocessing workflow adopted during the internship. Data extracted from operational systems first underwent cleaning, deduplication, and temporal filtering procedures to remove inconsistent observations and retain periods with sufficient historical coverage. Shipment-level records were then aggregated into weekly series, organized according to the hierarchical structures considered in this study, enriched with explanatory features, and transformed into representations compatible with statistical, machine learning, and neural forecasting models.

2.4.1 Weekly Aggregation

The original data is available at the shipment level, where each observation corresponds to a deposited international item. Since the objective of the internship is to forecast aggregated mail flows, the transactional data was resampled into weekly periods.

The choice of a weekly aggregation level was motivated by several considerations. First, weekly aggregation reduces the noise associated with day-to-day operational variability. Second, it better aligns with medium-term planning activities such as transport scheduling and capacity allocation. Finally, it allows the capture of annual seasonal effects while maintaining a sufficiently large number of observations for model estimation.

For each week, several aggregated indicators were computed, including:

- Total shipped weight;
- Number of shipments;
- Mean shipment weight;
- Standard deviation of shipment weight.

The total shipped weight was selected as the primary forecasting target, while the remaining indicators were retained for exploratory analyses and potential future extensions.

2.4.2 Hierarchical Structure Construction

One characteristic of international mail flows is their natural hierarchical organization. The same observations can be analyzed at different levels of aggregation, ranging from a national overview to detailed flows associated with specific deposit centers and destination countries.

In this study, a two-dimensional hierarchy was defined using:

- Deposit center (**Centre_Agence_depot**);
- Destination country (**Destination**).

An additional aggregation level corresponding to the national total was introduced to represent the root node of the hierarchy.

The hierarchy considered in this work is illustrated in Figure 2.5.

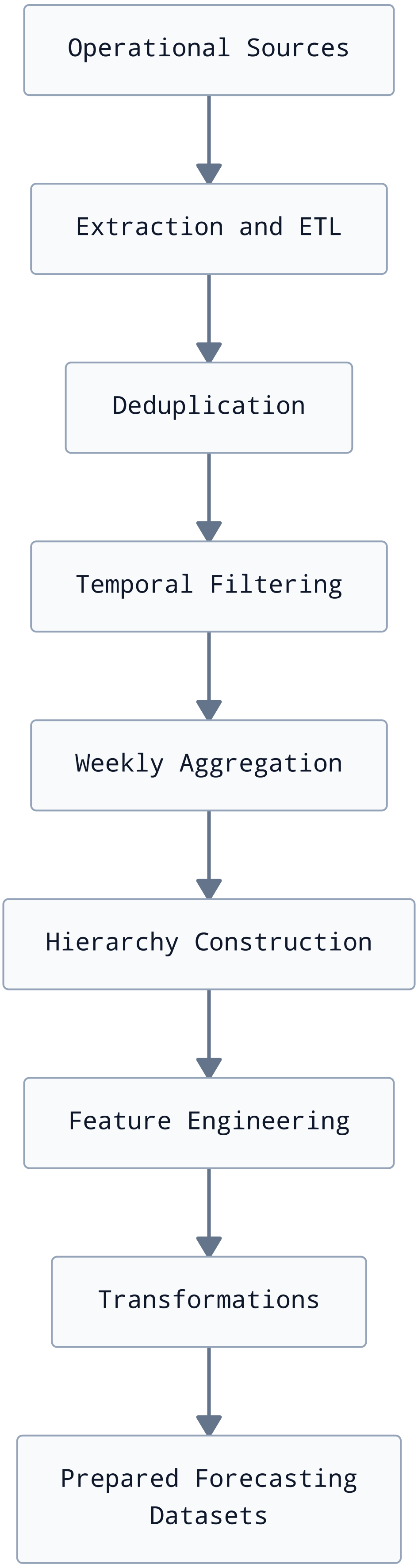

Figure 2.4 - Data preparation pipeline implemented for international mail forecasting.

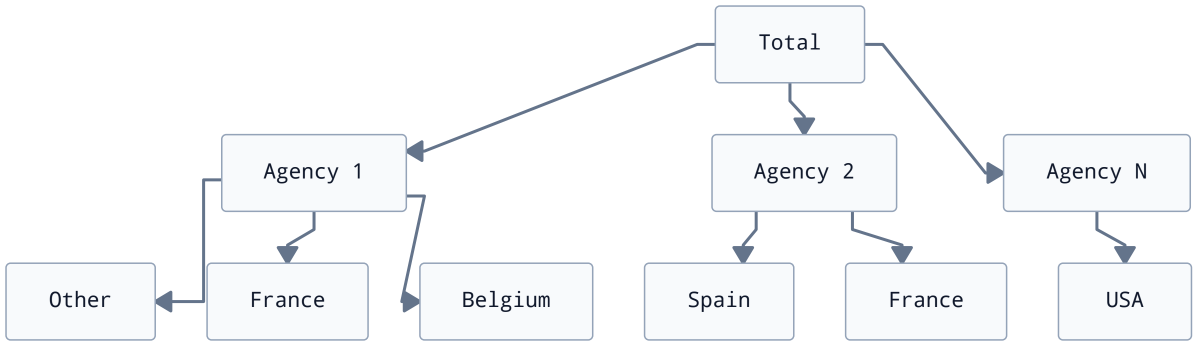

Figure 2.5 - Hierarchical organization adopted for international mail forecasting.

Figure 2.5 presents the aggregation structure retained for forecasting. Forecasts are initially generated at the root level and subsequently reconciled to ensure consistency across all hierarchy levels. This organization enables both global analyses and detailed investigations of specific origin-destination combinations.

The hierarchy was constructed using the aggregation utilities provided by the Hierarchical-Forecast library [13]. This process produces three essential objects:

- A dataset containing observations for all hierarchy levels;
- A summation matrix describing aggregation relationships;
- Metadata identifying each hierarchical level.

These structures are later used during the reconciliation stage described in Section 2.7.

2.4.3 Feature Engineering

Feature engineering plays a central role for machine learning and deep learning forecasting models. Unlike traditional statistical methods, these approaches generally require explicit explanatory variables to capture temporal dependencies.

Two categories of features were considered in this study: shared features used by multiple model families and model-specific features.

2.4.3.1 Shared Features

A deterministic trend component was generated using utilities from the UtilsForecast package [14]. This variable was incorporated into both machine learning and neural forecasting models.

The trend feature allows models to account for gradual changes in international mail demand that may not be fully captured through lagged observations alone.

2.4.3.2 Features for Machine Learning Models

Machine learning models were trained using lag-based supervised learning representations.

The following lag variables were considered:

- Lag 1;

- Lag 2;
- Lag 3;
- Lag 4;
- Seasonal lag corresponding to an approximate Hijri year.

The inclusion of a seasonal lag was motivated by preliminary analyses suggesting that international mail flows exhibit recurring patterns associated with religious events and seasonal demand cycles.

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- Lag 2;
- Lag 3;
- Lag 4;
- Seasonal lag corresponding to an approximate Hijri year.

The inclusion of a seasonal lag was motivated by preliminary analyses suggesting that international mail flows exhibit recurring patterns associated with religious events and seasonal demand cycles.

2.4.4 Transformations

Several forecasting models benefit from target transformations that stabilize variance and reduce the influence of extreme observations.

In this internship, a logarithmic transformation based on the function $ln(1 + x)$ was applied to the target variable before model fitting.

The transformed variable is defined as:

$$y'_t = ln(1 + y_t)$$

where y_t denotes the observed mail flow at time t.

Predictions produced in the transformed space were subsequently mapped back to the original scale using the inverse transformation.

This approach improves numerical stability and mitigates the effect of exceptionally large shipment volumes observed during peak periods.

2.5 Modeling Strategy

The forecasting system developed during this internship was designed around a modular architecture allowing several modeling paradigms to be evaluated under comparable conditions. This organization facilitates benchmarking while maintaining a common preprocessing and evaluation pipeline.

The overall structure of the modeling layer is summarized in Figure 2.6.

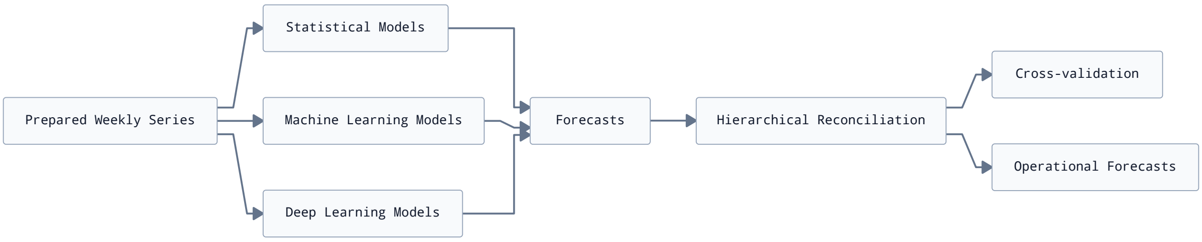

Figure 2.6 - Global architecture of the forecasting layer.

Figure 2.6 illustrates the organization adopted for model development. The prepared time series are provided to three complementary families of forecasting methods. The generated forecasts are subsequently reconciled across hierarchical levels and assessed using a common evaluation framework. This approach allows performance comparisons to focus on differences in modeling assumptions rather than discrepancies introduced during preprocessing.

The forecasting methods evaluated during this internship are summarized in Table 2.1.

Table 2.1 - Forecasting models evaluated during the internship.

Model	Family	Description
Seasonal Naive	Baseline	Seasonal benchmark based on historical repetition
AutoETS	State-space	Automatic exponential smoothing state-space model
AutoCES	State-space	Complex exponential smoothing model
AutoARIMA	Autoregressive	Automatic seasonal autoregressive integrated moving average model
AutoTBATS	Autoregressive	Trigonometric decomposition model for complex seasonal patterns
AutoTheta	Decomposition	Theta decomposition forecasting approach
AutoMFLES	Ensemble smoothing	Multi-feature local ensemble smoothing model
LinearRegression	Linear ML	Ordinary least squares regression
Ridge	Linear ML	L2-regularized linear regression
Lasso	Linear ML	L1-regularized linear regression
ElasticNet	Linear ML	Combined L1/L2 regularization
ARDRegression	Bayesian ML	Automatic relevance determination regression
TweedieRegressor	Generalized linear model	Tweedie-distributed regression
HistGradientBoostingRegressor	Boosting ML	Histogram-based gradient boosting trees
LGBMRegressor	Boosting ML	Light Gradient Boosting Machine
XGBRegressor	Boosting ML	Extreme Gradient Boosting

Model	Family	Description
CatBoostRegressor	Boosting ML	Gradient boosting with ordered boosting
NHITS	Deep Learning	Neural hierarchical interpolation model
BiTCN	Deep Learning	Bidirectional temporal convolutional network

Table 2.1 illustrates the diversity of forecasting paradigms considered in this work. In addition to traditional statistical methods, several machine learning and deep learning approaches were investigated to assess their suitability for modeling international postal flows.

2.5.1 Baseline Models

A baseline model is necessary to determine whether more sophisticated approaches provide a meaningful improvement in predictive accuracy. In time-series forecasting, simple benchmarks are often difficult to outperform, particularly when seasonality is present [1].

Among the models summarized in Table 2.1, the Seasonal Naive model was selected as the baseline forecasting approach.

The Seasonal Naive approach assumes that future observations repeat the behavior observed during the previous seasonal cycle. Given the characteristics identified during exploratory analyses, a seasonal period corresponding approximately to one Hijri year was adopted.

Formally, the Seasonal Naive forecast is expressed as:

$$\hat{y}_{t+h} = y_{t+h-m}$$

where:

- m denotes the seasonal period;
- h is the forecasting horizon.

This model serves two purposes. First, it establishes a minimum acceptable level of forecasting performance. Second, it is used as a reference in relative evaluation metrics such as Relative Mean Absolute Error (RMAE).

2.5.2 Statistical Forecasting Models

Statistical forecasting methods remain widely used in industrial applications due to their robustness, interpretability, and relatively low computational requirements [1].

Several automated statistical models available through the StatsForecast framework [15] were considered in this study.

2.5.2.1 State-space models

State-space methods represent observations through latent components describing the evolution of level, trend, and seasonality. Their recursive estimation procedures allow model parameters to adapt efficiently to changing dynamics while maintaining relatively low computational costs.

2.5.2.1.1 AutoETS

The Exponential Smoothing State Space (ETS) framework models observations as combinations of error, trend, and seasonal components estimated within a state-space formulation [16]. The AutoETS implementation automatically searches among admissible model configurations and selects the most appropriate specification according to information criteria.

ETS models are widely regarded as strong benchmarks in business forecasting applications because they provide interpretable decompositions while remaining computationally efficient. Their ability to accommodate trend and seasonal variations motivated their inclusion in this comparative study.

2.5.2.1.2 AutoCES

Complex Exponential Smoothing (CES) extends traditional exponential smoothing methods by representing cyclical behavior through complex-valued states [17].

Compared with conventional ETS formulations, CES can better accommodate oscillatory dynamics and evolving seasonal structures. Although it remains less widely adopted in industrial forecasting systems, its recent development and promising empirical performance motivated its inclusion as an exploratory statistical model.

2.5.2.2 Autoregressive models

Autoregressive approaches explain future observations as functions of lagged values and previous forecast errors. These methods explicitly characterize temporal dependence structures and remain among the most widely used forecasting techniques.

2.5.2.2.1 AutoARIMA

AutoARIMA automatically identifies suitable autoregressive, differencing, and moving-average orders through an optimization procedure based on information criteria [18], [19].

By incorporating seasonal components, AutoARIMA can capture persistent temporal dependencies and recurring patterns observed in historical observations. Due to its strong theoretical foundation and widespread industrial use, it was included as a reference statistical approach.

2.5.2.2.2 AutoTBATS

TBATS combines trigonometric seasonal representations, Box-Cox transformations, autoregressive error corrections, and damped trend components within a unified framework [20].

The method was originally designed to model complex and multiple seasonal structures. Although the present study primarily considered a single annual seasonal pattern approximated by the Hijri calendar, AutoTBATS was retained because of its flexibility in representing non-standard seasonal behaviors.

2.5.2.3 Decomposition models

Decomposition approaches attempt to separate a time series into simpler latent components before extrapolating their future evolution.

2.5.2.3.1 AutoTheta

The Theta method decomposes a time series into modified versions referred to as Theta lines, which are subsequently extrapolated and recombined to generate forecasts [21].

The method achieved notable success in forecasting competitions and is often recognized for providing competitive predictive performance despite its conceptual simplicity. AutoTheta was therefore included as a robust decomposition-based benchmark.

2.5.2.4 Ensemble smoothing models

Ensemble smoothing methods attempt to combine the robustness of classical smoothing procedures with greater flexibility in adapting to local changes in the observed dynamics.

2.5.2.4.1 AutoMFLES

AutoMFLES is a recently proposed forecasting approach that combines smoothing techniques with local feature extraction mechanisms to improve responsiveness to changing patterns [22].

Given its relatively recent introduction and limited evaluation in postal forecasting contexts, AutoMFLES was incorporated primarily as an exploratory candidate aimed at assessing whether more flexible smoothing strategies could provide additional forecasting gains.

2.5.3 Machine Learning Forecasting Models

Unlike statistical forecasting methods, machine learning approaches do not rely on explicit assumptions regarding the underlying stochastic process generating the observations. Instead, they learn relationships between engineered explanatory variables and future values directly from historical data.

Machine learning models were implemented using the MLForecast framework [23], which provides a unified interface for feature generation, model training, and forecasting. All models were trained using the feature engineering pipeline described in Section 2.4.3, including lagged observations and trend indicators.

The machine learning models considered in this study can be grouped into linear models, Bayesian regression models, generalized linear models, and gradient boosting approaches.

2.5.3.1 Linear models

Linear models remain attractive forecasting methods because they are computationally efficient, highly interpretable, and often provide surprisingly competitive performance when combined with carefully engineered features.

2.5.3.1.1 LinearRegression

LinearRegression estimates model parameters using ordinary least squares (OLS), minimizing the residual sum of squares between observed and predicted values. Despite its simplicity, linear regression often constitutes a strong baseline in forecasting problems where temporal dependencies can be effectively represented through lagged variables and rolling statistics.

Its inclusion in this study aimed to assess the predictive value of the engineered features independently of more sophisticated regularization or nonlinear modeling techniques.

2.5.3.1.2 Ridge

Ridge regression extends ordinary least squares by introducing an L2 regularization penalty that shrinks regression coefficients toward zero, thereby reducing variance and mitigating overfitting. [24]

This regularization mechanism is particularly useful when explanatory variables exhibit strong collinearity, as is frequently the case in time series forecasting applications involving multiple lagged features and moving averages.

Among the machine learning approaches evaluated during this internship, Ridge regression demonstrated consistently strong predictive performance and emerged as one of the most competitive models within its category.

2.5.3.1.3 Lasso

Lasso regression incorporates an L1 regularization penalty that encourages sparse solutions by driving some regression coefficients exactly to zero. [25]

This property allows Lasso to perform implicit feature selection, potentially improving model interpretability and reducing sensitivity to irrelevant predictors. However, its ability to discard explanatory variables may also limit predictive performance when numerous correlated temporal features contribute useful information.

2.5.3.1.4 ElasticNet

ElasticNet combines L1 and L2 regularization penalties within a unified framework, seeking to balance the sparsity induced by Lasso with the stability provided by Ridge regression. [26]

This hybrid formulation can be advantageous when dealing with high-dimensional feature spaces containing groups of correlated variables. Its inclusion in the experimental framework aimed to evaluate whether intermediate regularization strategies could improve forecasting accuracy.

2.5.3.2 Bayesian and generalized linear models

Bayesian and generalized linear approaches provide additional flexibility by introducing probabilistic assumptions regarding parameter estimation or response distributions.

2.5.3.2.1 ARDRegression

Automatic Relevance Determination (ARD) regression adopts a Bayesian perspective by estimating individual prior distributions for regression coefficients. Through an iterative optimization procedure, irrelevant predictors receive stronger shrinkage, allowing the model to automatically identify informative explanatory variables. [27]

ARD regression was included to investigate whether Bayesian regularization techniques could improve robustness when handling large collections of engineered temporal features.

2.5.3.2.2 TweedieRegressor

The TweedieRegressor belongs to the family of generalized linear models and assumes that observations follow a Tweedie distribution.

Such distributions are often appropriate for modeling positive-valued or zero-inflated quantities frequently encountered in insurance, finance, and demand forecasting applications. Since international mail volumes are non-negative by construction, the TweedieRegressor was considered a potentially suitable alternative to classical linear regression approaches.

2.5.3.3 Gradient boosting models

Gradient boosting algorithms construct ensembles of decision trees in a sequential manner, where each newly added tree attempts to correct the residual errors produced by previous iterations.

These methods are capable of capturing complex nonlinear interactions between explanatory variables and have become widely adopted in forecasting competitions and industrial machine learning applications.

2.5.3.3.1 HistGradientBoostingRegressor

HistGradientBoostingRegressor is a histogram-based implementation of gradient boosting available in Scikit-learn. By discretizing continuous variables into bins prior to tree construction, it significantly reduces computational costs while maintaining competitive predictive performance. [28]

Its inclusion provided a computationally efficient boosting baseline within the machine learning family.

2.5.3.3.2 LGBMRegressor

LightGBM is a highly optimized gradient boosting framework designed to improve training speed and scalability through histogram-based learning and leaf-wise tree growth strategies. [29]

LightGBM has demonstrated strong performance in numerous tabular machine learning tasks and was therefore considered a natural candidate for evaluating nonlinear relationships within the engineered feature space.

2.5.3.3.3 XGBRegressor

XGBoost is an implementation of gradient boosting that incorporates regularization mechanisms, efficient tree construction algorithms, and parallelized optimization procedures. [30]

Due to its widespread success in predictive analytics and forecasting competitions, XGBoost was included as a representative state-of-the-art boosting algorithm.

2.5.3.3.4 CatBoostRegressor

CatBoost is a gradient boosting framework originally developed to improve the treatment of categorical variables through ordered boosting procedures. [31]

Although the forecasting problem considered in this internship primarily relied on numerical features, CatBoost remains highly competitive on structured datasets and demonstrated strong predictive performance among the machine learning models evaluated.

2.5.4 Deep Learning Models

Deep learning approaches have recently attracted considerable interest in time-series forecasting due to their ability to learn complex nonlinear relationships directly from historical observations [1]. Unlike classical statistical models and most machine learning methods, neural forecasting models can automatically discover hierarchical temporal representations without relying heavily on manually engineered features.

The neural models evaluated in this study were implemented using the NeuralForecast framework [32]. In contrast to the machine learning models presented previously, only a limited set of covariates was supplied to the neural architectures, as these models are designed to learn temporal dependencies directly from sequential data.

Deep learning models are generally characterized by a larger number of trainable parameters and consequently require substantial amounts of data to reach their full predictive potential. During preliminary experiments, their performance was found to be highly dependent on the amount of training data available. This behavior was particularly evident in the temporal evolution of cross-validation scores, where neural models initially underperformed simpler approaches but progressively became more competitive as additional observations were incorporated into the training windows.

Only two neural architectures were retained for the final experimental evaluation: NHITS and BiTCN.

2.5.4.1 NHITS

NHITS (Neural Hierarchical Interpolation for Time Series Forecasting) is a deep neural architecture specifically designed for long-horizon forecasting problems [33]. The model extends the N-BEATS[34] family by introducing hierarchical interpolation mechanisms that improve computational efficiency while preserving the ability to model complex temporal dynamics.

NHITS decomposes the forecasting task into several blocks operating at different temporal resolutions. Lower-frequency components capture long-term trends and seasonal patterns, while higher-frequency components focus on short-term fluctuations. The outputs of these blocks are subsequently combined to produce the final forecast.

Its ability to model multiple temporal scales motivated its inclusion in this study, particularly given the coexistence of long-term trends and recurring seasonal effects observed in international postal traffic.

2.5.4.2 BiTCN

BiTCN is a forecasting architecture derived from Temporal Convolutional Networks (TCNs) [35] and implemented within the NeuralForecast ecosystem [32]. The model employs dilated convolutional layers to efficiently capture long-range temporal dependencies while maintaining parallelizable training procedures. [36]

Unlike recurrent neural networks, convolutional architectures process entire sequences simultaneously, allowing significantly faster training and inference. The bidirectional structure adopted by BiTCN enables information to be extracted from multiple receptive fields, facilitating the representation of complex temporal interactions.

Among all forecasting models evaluated during this internship, BiTCN achieved the best predictive performance for the short forecasting horizon considered in the operational use case ($h = 4$ weeks). However, additional experiments conducted on extended horizons revealed a noticeable deterioration in neural forecasting performance, suggesting that the advantages offered by deep architectures may be more pronounced for short-term operational forecasting scenarios than for long-range extrapolation tasks under the available data regime.

2.5.5 Hierarchical Forecasting and Forecast Reconciliation

As discussed in Section 2.4.2, outgoing international mail flows naturally exhibit a hierarchical structure. The same observations can be aggregated at different levels, including the overall international flow, deposit centers, destination countries, or combinations of these dimensions.

Forecasting each series independently generally produces incoherent results. For example, the sum of forecasts generated for individual deposit centers may differ from the forecast obtained directly for the national total. Such inconsistencies complicate operational planning and reduce the interpretability of forecasting outputs.

Hierarchical forecasting addresses this issue by combining forecasting and reconciliation techniques to ensure aggregation coherence across all levels of the hierarchy [1].

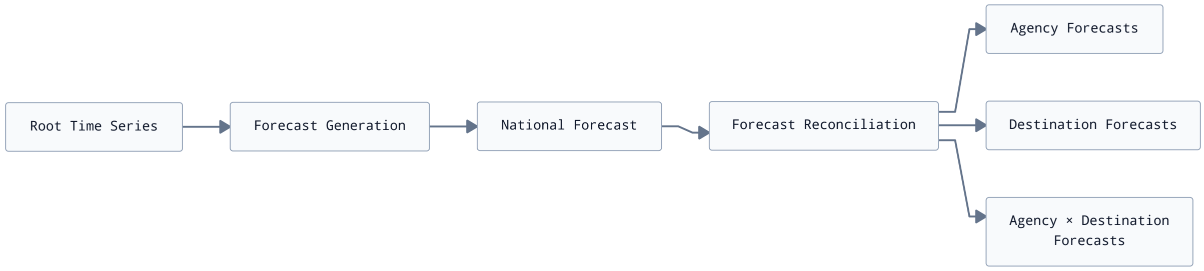

Figure 2.7 - Forecast reconciliation process adopted during the internship.

Figure 2.7 presents the strategy implemented in this work. Forecasts are first generated at the aggregate level and subsequently distributed across lower hierarchical levels through a reconciliation procedure. This approach ensures consistency between detailed forecasts and the overall prediction.

The hierarchy considered in this study is composed of two dimensions:

- Deposit center;
- Destination country.

An additional node representing the total international outgoing flow was introduced as the root of the hierarchy.

The aggregation structures required for reconciliation were constructed using the utilities provided by the HierarchicalForecast package [13]. This process generates three main objects:

- A dataset containing observations for all aggregation levels;
- A summation matrix describing hierarchical relationships;
- Metadata identifying each level of the hierarchy.

The main reconciliation approaches proposed in the literature are summarized in Table 2.2.

Table 2.2 - Common hierarchical reconciliation approaches.

Method	Principle
Bottom-Up	Aggregates forecasts generated at the most detailed level.
Top-Down	Distributes aggregate forecasts to lower levels using allocation proportions.
MinTrace [37]	Produces coherent forecasts while minimizing reconciliation errors.

Table 2.2 shows that several reconciliation strategies can be considered. Their suitability depends on the forecasting context, the availability of historical information, and the reliability of forecasts at different aggregation levels.

In this internship, the Top-Down reconciliation method based on historical average proportions was selected. This approach was considered appropriate because the forecasting models were trained exclusively on the aggregate international series.

The reconciliation process consists of allocating the forecast generated at the root level among subordinate nodes according to proportions estimated from historical observations.

If

$$\hat{y}_{\text{Total},t+h}$$

denotes the aggregate forecast and p_i the historical proportion associated with node i, the reconciled forecast is computed as:

$$\hat{y}_{i,t+h} = p_i \hat{y}_{\text{Total},t+h}$$

where:

- p_i is the average historical contribution of node i;
- $\hat{y}_{\text{Total},t+h}$ is the forecast produced at the aggregate level.

This strategy allows coherent forecasts to be generated for all combinations of deposit centers and destination countries while avoiding the need to train a separate forecasting model for each individual series.

Although more sophisticated reconciliation techniques such as MinTrace could potentially improve predictive accuracy, the Top-Down method offered a good compromise between implementation complexity, computational cost, and interpretability within the context of this internship.

2.6 Evaluation Methodology

The final stage of the modeling phase in CRISP-DM consists of assessing the predictive performance of the candidate models and selecting the approaches that are the most suitable for the target application [12]. In the context of time-series forecasting, model evaluation requires methodologies that preserve the temporal ordering of observations and mimic realistic forecasting scenarios.

This section presents the validation strategy adopted during the internship, the forecasting horizon considered, and the evaluation metrics used to compare candidate models.

2.6.1 Forecasting Horizon

The forecasting horizon adopted in this study is four weeks.

The choice of a short-term horizon was motivated by operational considerations. Discussions with the internship supervisors indicated that forecasts covering approximately one month are sufficiently long to support transportation planning, workload anticipation, and resource allocation decisions, while remaining within a time range where forecasting uncertainty is still manageable.

Consequently,

$$h = 4$$

was selected as the prediction horizon for all candidate models.

2.6.2 Rolling-Origin Cross-Validation

Model evaluation was conducted using a rolling-origin cross-validation procedure inspired by the approach described in [1] and supported by empirical evidence on the validity of time-series cross-validation for autoregressive forecasting tasks [38].

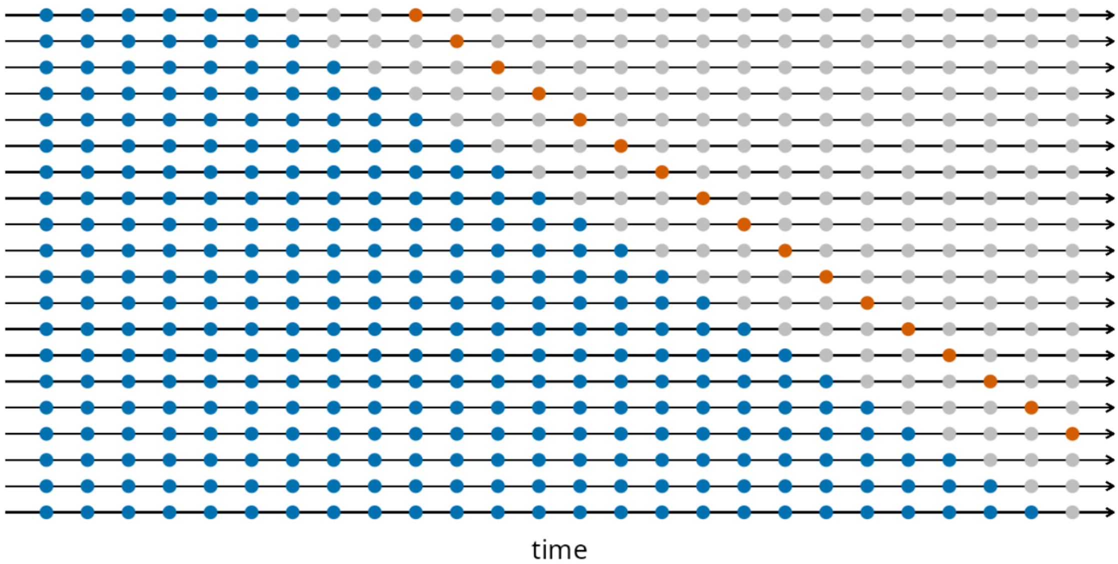

Figure 2.8 - Rolling-origin cross-validation procedure adopted during model evaluation. (figure adapted from [1])

Figure 2.8 illustrates the rolling-origin cross-validation strategy implemented during this internship. Instead of randomly splitting observations into training and testing subsets, successive forecasting exercises are performed by progressively moving the forecasting origin forward through time. This methodology preserves the chronological structure of the data and provides a more realistic assessment of forecasting performance. For each forecasting exercise, models were trained on the observations available up to a given cutoff date and then used to predict the following four weeks.

The process was repeated over ten evaluation windows.

The main validation parameters are summarized in Table 2.3.

Table 2.3 - Parameters used for cross-validation.

Parameter	Value
Forecast horizon	4 weeks
Number of windows	10
Window step	10 weeks
Evaluation strategy	Rolling-origin cross-validation

Table 2.3 summarizes the configuration adopted for model evaluation. Using multiple validation windows makes it possible to assess model robustness under different demand conditions and reduces the risk of selecting a model that performs well only during a specific period.

2.6.3 Evaluation Metrics

Several complementary performance indicators were used to compare candidate forecasting models.

Using multiple metrics is recommended because no single indicator can fully characterize forecasting quality [1].

The metrics considered during this internship are described in Table 2.4.

Table 2.4 - Evaluation metrics used during model comparison.

Metric	Interpretation
MASE	Mean Absolute Scaled Error relative to an in-sample seasonal benchmark
RMAE	Relative Mean Absolute Error compared with the Seasonal Naive model
ND	Normalized Deviation measuring the magnitude of forecast errors
SPIS	Scaled interval score evaluating probabilistic forecasts

Table 2.4 presents the performance indicators retained in this study. The use of relative metrics such as RMAE facilitates comparisons against the Seasonal Naive baseline, while interval-based measures provide information about forecast uncertainty.

2.6.4 Baseline Comparison

The Seasonal Naive model was used as a reference model throughout the evaluation process.

This baseline was selected because preliminary analyses suggested the presence of recurring annual patterns in international mail flows. The seasonal period adopted corresponds approximately to one Hijri year, represented by fifty-one weekly observations.

Relative metrics, particularly RMAE, were computed with respect to this baseline. Values below one indicate an improvement over the Seasonal Naive forecast, whereas values above one suggest inferior predictive performance.

2.6.5 Prediction Intervals

In addition to point forecasts, uncertainty estimation was considered whenever supported by the forecasting framework.

For statistical models implemented through StatsForecast, conformal prediction intervals were investigated. In practice, only the MFLES model was able to integrate conformal intervals reliably within the experimental setting adopted during this internship.

The confidence levels considered in preliminary experiments were 80% and 95%.

Although probabilistic forecasting was not the primary objective of the internship, incorporating uncertainty estimates can provide useful information for operational planning, especially during periods characterized by strong demand fluctuations.

2.6.6 2.7.3 Model Comparison Strategy

The objective of the model comparison procedure is to identify forecasting approaches that provide accurate, stable, and operationally relevant predictions for outgoing international mail flows. Since no single performance indicator is sufficient to characterize forecasting quality, candidate models are evaluated according to multiple complementary criteria.

2.6.6.1 Ranking Procedure

Candidate models are evaluated using the rolling-origin cross-validation framework described in Section 2.6.2. For each validation window, forecasting errors are computed using the metrics presented in Section 2.6.3.

Model comparison is primarily based on the average value of each metric over all cross-validation windows. Particular attention is given to the Relative Mean Absolute Error (RMAE), which measures the performance of a model relative to the SeasonalNaive baseline and provides an intuitive interpretation of the forecasting gain obtained by more sophisticated approaches.

The comparison process includes statistical models, machine learning methods, deep learning architectures, and ensemble forecasts. Models are ranked according to their average performance while also considering the consistency of their behavior across validation windows.

2.6.6.2 Ensemble Construction

In addition to evaluating individual models, ensemble forecasts are investigated. Rather than combining all available forecasting approaches, ensembles are constructed within each methodological family by selecting two models exhibiting strong predictive performances and sufficiently different modeling assumptions.

Table 2.5 summarizes the composition of the ensemble forecasts considered in this study.

Table 2.5 - Composition of the ensemble forecasting models.

Ensemble	Components
stat_ensemble	AutoARIMA + CES
ml_ensemble	Ridge + CatBoostRegressor
neural_ensemble	BiTCN + NHITS

As shown in Table 2.5, ensemble forecasts are generated by averaging the predictions produced by their constituent models. This strategy aims to reduce the influence of individual model biases, improve forecast stability, and exploit complementary forecasting behaviors while maintaining a relatively simple implementation.

2.6.6.3 Final Model Selection Criteria

The selection of the final forecasting approach is not solely based on minimizing a single error metric. Several aspects are jointly considered during the decision process, including:

- Average forecasting accuracy;
- Stability across validation windows;
- Ability to benefit from larger training datasets;
- Compatibility with hierarchical reconciliation procedures;
- Ease of integration into the proposed forecasting pipeline;
- Suitability for operational deployment and visualization.

The combination of these criteria makes it possible to identify models that not only achieve competitive predictive performances but also satisfy the practical constraints associated with operational use within Barid Al-Maghrib.

The evaluation framework described in this section was applied consistently to all forecasting approaches. The resulting performance comparisons and model selection process are presented in Chapter 3.

2.7 Operationalization and Forecast Delivery Design

The operationalization phase defines how the forecasting system is executed in practice and how model outputs are transformed into usable operational artifacts. In the CRISP-DM framework, this stage bridges the gap between modeling results and their integration into decision-support workflows [12].

In the context of this internship, operationalization is not limited to deployment in a production environment, but rather focuses on the design of a reproducible pipeline capable of generating forecasts, reconciling hierarchical outputs, and producing structured datasets for analysis and visualization.

2.7.1 Forecast Generation Pipeline

The forecasting pipeline is organized as a multi-stage process that transforms preprocessed time series into future predictions.

The process is composed of the following steps:

- preparation of hierarchical time series inputs;
- generation of root-level forecasts;
- projection of exogenous features when required;
- application of trained forecasting models;
- generation of multi-horizon predictions ($h = 4$ for evaluation, $h = 100$ for operational planning).

This pipeline is implemented in a modular way in order to support multiple model families, including statistical, machine learning, and neural forecasting approaches.

The root-level forecasting strategy is consistent across all model families, ensuring comparability of results before hierarchical reconciliation is applied.

2.7.2 Hierarchical Reconciliation Output

Raw forecasts generated by the different models are not directly suitable for operational use due to inconsistencies across hierarchical levels. In particular, independently generated forecasts do not satisfy aggregation constraints across deposit centers and destination groups.

To address this issue, a hierarchical reconciliation step is applied after forecast generation.

The reconciliation procedure follows a top-down allocation strategy based on historical proportions[39]. Let:

$$\hat{y}_{i,t+h} = p_i \hat{y}_{\text{Total},t+h}$$

where:

- $\hat{y}_{\text{Total},t+h}$ represents the forecast at the aggregate level;
- p_i represents the historical proportional contribution of node i;
- $\hat{y}_{i,t+h}$ is the reconciled forecast for node i.

This procedure ensures structural consistency across all levels of the hierarchy while preserving the dynamics captured at the aggregate forecasting level.

The implementation is based on the HierarchicalForecast library, which provides utilities for defining hierarchical structures and applying reconciliation methods [13].

Figure 2.9 illustrates the transformation from raw model outputs to coherent hierarchical forecasts.

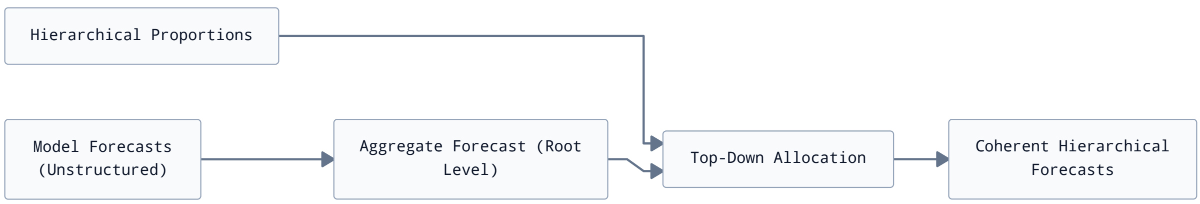

Figure 2.9 - Hierarchical reconciliation applied to model outputs.

Figure 2.9 shows that reconciliation is applied as a post-processing step after forecast generation. This ensures that all hierarchical constraints are satisfied while maintaining consistency with the root-level predictive signal.

2.7.3 Output Data Schema and Export Layer

Once forecasts are reconciled, results are structured into a unified dataset suitable for analysis, evaluation, and visualization.

The final dataset includes:

- time index (ds);
- hierarchical identifiers (deposit center, destination, or aggregated node);

- observed values (for evaluation periods);
- predicted values (for all models);
- reconciled forecasts;
- model metadata.

This structured format allows direct integration with evaluation routines as well as visualization tools.

2.7.4 Interactive Visualization System (Altair Export Layer)

An interactive visualization layer was developed to support exploratory analysis and operational interpretation of forecasts.

The system is based on Altair and provides dynamic filtering capabilities across hierarchical dimensions.

The visualization pipeline is structured as follows:

- conversion of Polars dataframes to Pandas for compatibility with Altair;
- generation of time-series plots for observed and forecasted values;
- synchronization of multiple model outputs within a single view;
- integration of dropdown filters for hierarchical navigation;
- rendering of interactive web-based charts.

The visualization layer was designed to expose reconciled forecasts generated by all evaluated forecasting models. This design enables analysts to compare forecasting behaviors interactively and assess the impact of model selection at different aggregation levels.

Rather than being restricted to a single forecasting approach, the visualization system acts as a generic interface over the forecasting outputs produced during the evaluation phase.

Figure 2.10 illustrates the structure of the visualization system.

Figure 2.10 shows how reconciled forecasts are transformed into an interactive visualization. The system enables filtering across hierarchical levels while maintaining synchronization between observed values and multiple forecasting outputs.

Conclusion

This chapter presented the methodological framework adopted in this internship following a CRISP-DM-oriented structure adapted to the specific constraints of postal data forecasting.

The business understanding phase clarified the operational context of outgoing international mail flows and highlighted the need for short-term forecasting to support capacity planning and logistics optimization. The problem was formalized as a multi-step time-series forecasting task with a fixed horizon of four weeks.

The data understanding and preparation phases detailed the construction of a structured dataset from heterogeneous operational sources. Particular attention was given to the hierarchical nature of the data, requiring aggregation across multiple levels such as deposit centers and destination countries. Due to the absence of a dedicated analytical infrastructure, signif-

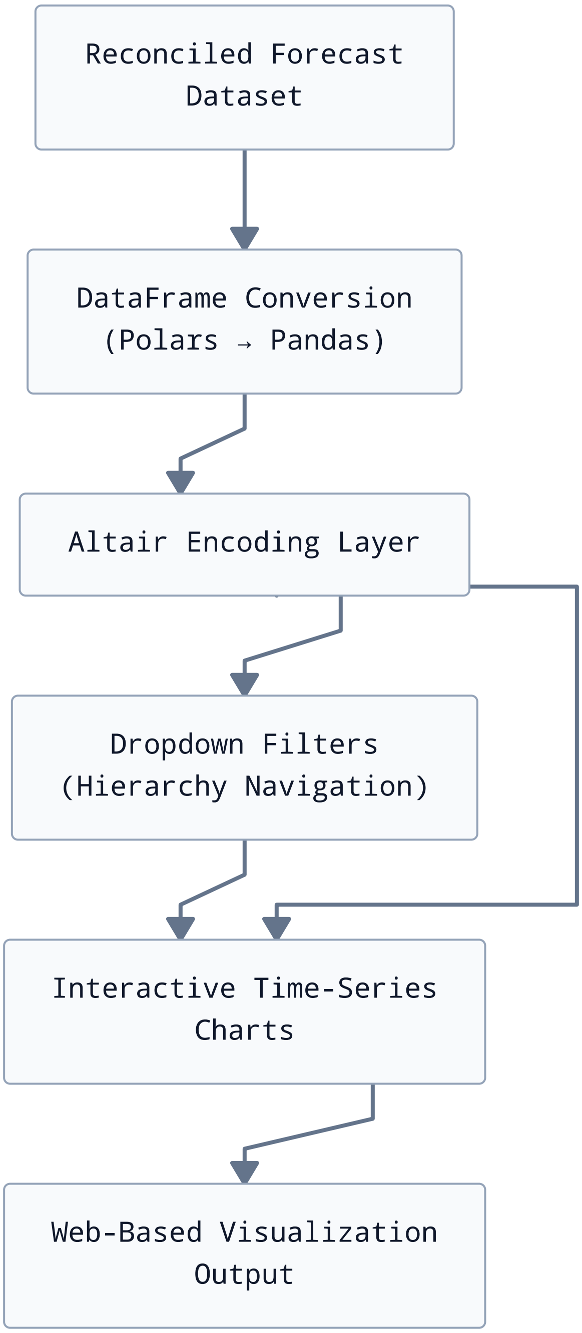

Figure 2.10 - Architecture of the interactive Altair visualization system.

icant effort was required to reconstruct consistent historical time series from operational systems.

The modeling strategy introduced a multi-paradigm forecasting framework combining statistical models, machine learning approaches, and deep learning architectures. All models were trained under a unified experimental protocol to ensure comparability. Hierarchical reconciliation was introduced to guarantee coherence between forecasts at different aggregation levels.

The evaluation methodology was based on rolling-origin cross-validation, ensuring a realistic simulation of operational forecasting conditions. Multiple complementary metrics were used to assess both accuracy and robustness of the candidate models.

Finally, the operationalization layer defined how forecasts are generated, reconciled, and structured into a unified output format. An interactive visualization system based on Altair was also designed to enable exploratory analysis and support interpretation of results.

The following chapter presents the implementation details of the system, including data engineering pipelines, model training procedures, and the integration of hierarchical forecasting and visualization components into a complete end-to-end workflow.

Chapter 3
Realization and Results

Introduction

This chapter presents the implementation of the forecasting system described in Chapter 2. It details how the methodological components were translated into a working pipeline for data extraction, processing, modeling, hierarchical reconciliation, and visualization.

Unlike the methodology chapter, which focused on design choices and theoretical justification, this chapter emphasizes practical implementation decisions, system integration, and experimental results obtained from the executed pipeline.

The implementation covers three main aspects:

- data engineering and construction of the analytical dataset;
- implementation of forecasting models across statistical, machine learning, and neural paradigms;
- deployment of hierarchical reconciliation and interactive visualization components.

3.1 System Architecture

The forecasting system is implemented as an end-to-end pipeline that connects raw operational postal data to final analytical outputs, including forecasts and interactive visualizations.

The architecture follows a layered design separating data ingestion, transformation, modeling, and presentation.

3.1.1 End-to-End Pipeline

The complete pipeline implemented during this internship is illustrated in Figure 3.1.

Figure 3.1 summarizes the full operational workflow implemented in this internship. Raw operational data is progressively transformed into structured hierarchical time series, enriched with engineered features, and processed by multiple forecasting families. The resulting predictions are reconciled and then evaluated and visualized.

This design ensures traceability between raw operational events and final forecasting outputs, which is essential in a production-constrained environment where data is not directly available in analytical form.

3.1.2 Technology Stack

The system was implemented using a combination of data engineering, machine learning, and visualization tools.

The main technologies used are summarized in Table 3.1.

Table 3.1 shows that the implementation relies on a modular ecosystem where each library addresses a specific layer of the pipeline. This separation of concerns allowed independent development of data engineering, modeling, and visualization components while maintaining interoperability through standardized data structures.

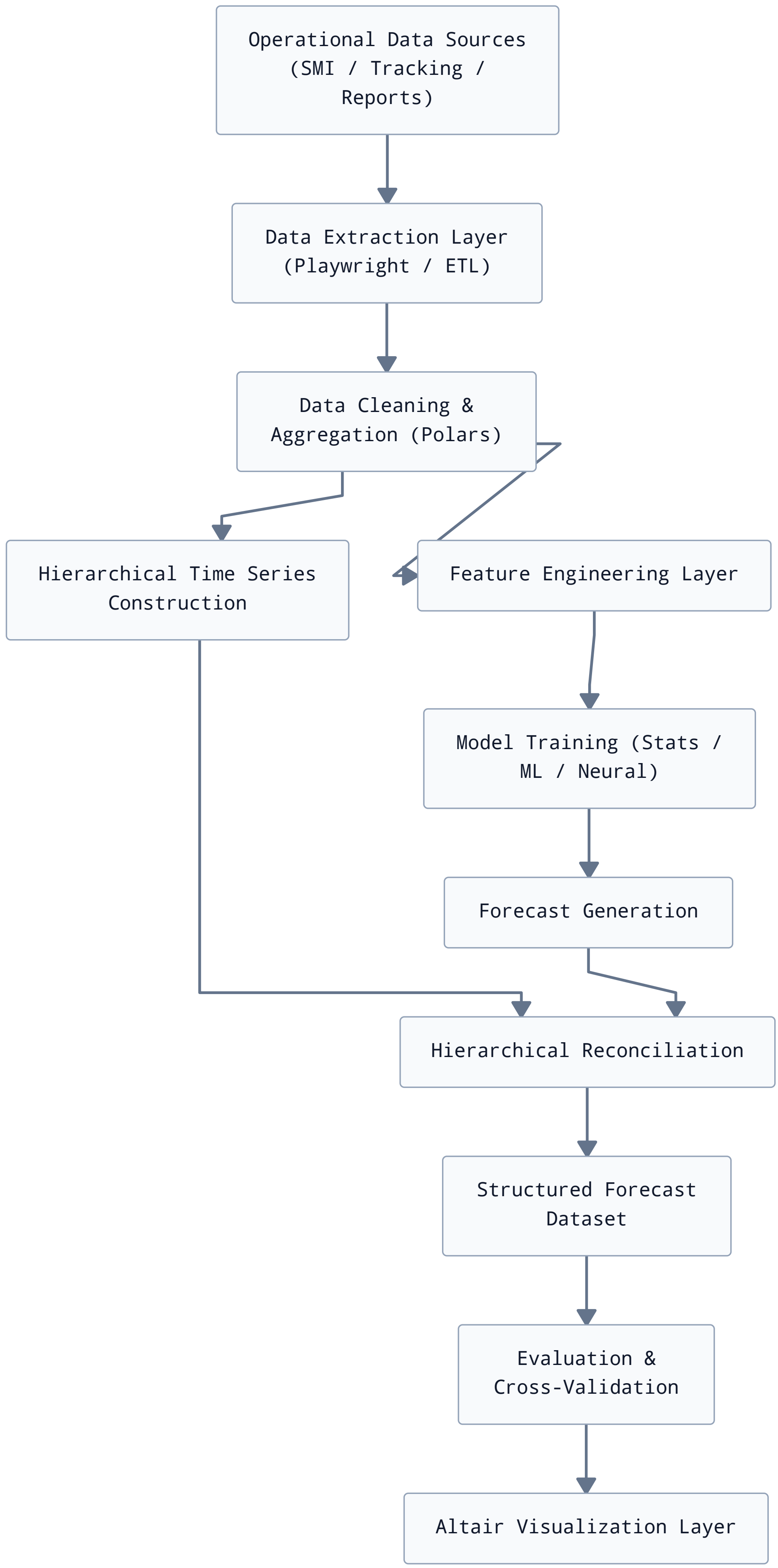

Figure 3.1 - End-to-end forecasting pipeline implemented during the internship.

Table 3.1 - Technology stack used in the forecasting system.

Component	Tools / Libraries
Data processing	Python, Polars, Pandas
Data extraction	Playwright
Statistical forecasting	StatsForecast
Machine learning forecasting	MLForecast, scikit-learn, XGBoost, LightGBM, CatBoost
Deep learning forecasting	NeuralForecast (NHITS, BiTCN)
Hierarchical forecasting	HierarchicalForecast
Visualization	Altair, VegaFusion
Evaluation	utilsforecast, statsmodels metrics

3.2 Data Engineering Implementation

This section details the implementation of the data engineering layer responsible for transforming raw operational postal data into structured datasets suitable for forecasting. It corresponds to the execution of the data preparation strategy described in Section 2.4.

The implementation addresses two main constraints:

- absence of a dedicated analytical database;
- reliance on operational interfaces designed for manual consultation rather than bulk extraction.

As a result, the data pipeline is built around automated extraction, intermediate storage, and repeated aggregation steps.

3.2.1 Data Extraction Layer

The raw data used in this internship originates from operational systems related to international mail processing (SMI and tracking interfaces). Since these systems do not provide a stable analytical API, data extraction was implemented using browser automation.

The extraction workflow is based on Playwright and simulates user interactions with internal web interfaces.

The process can be summarized as follows:

- authentication into operational interfaces;
- navigation to reporting or tracking pages;
- query execution over selected time ranges;
- extraction of tabular HTML content;
- conversion into structured formats.

The extraction mechanism is summarized in Figure 3.2.

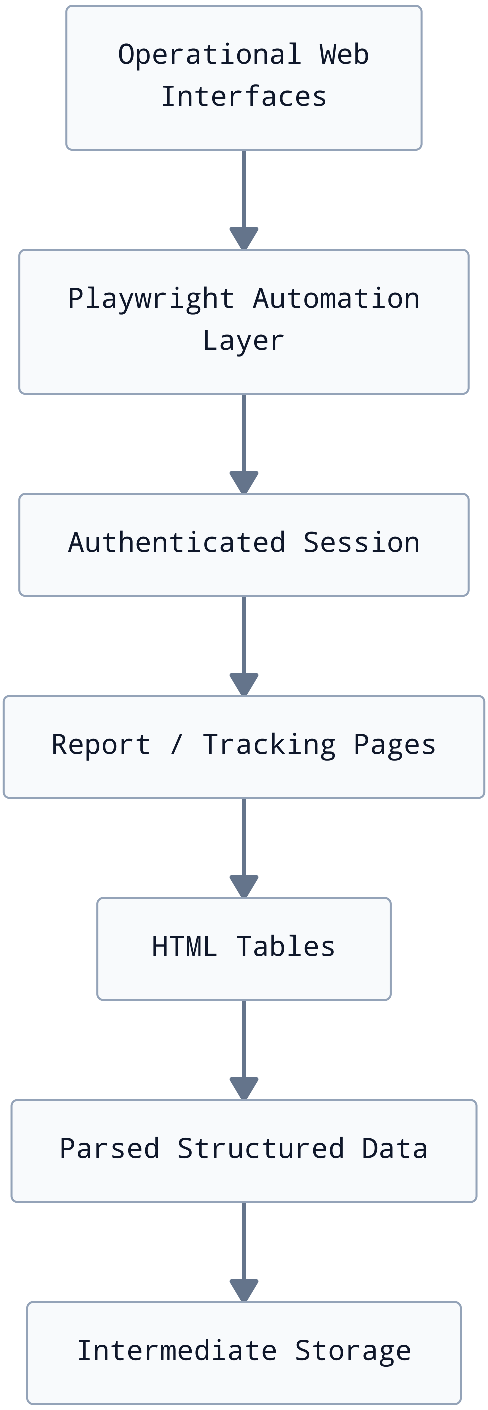

Figure 3.2 - Automated data extraction pipeline from operational systems.

Figure 3.2 illustrates the transformation of operational web interfaces into structured datasets. The use of browser automation ensures reproducibility of data extraction while compensating for the absence of a formal API.

3.2.2 Aggregation and Time-Series Construction

After extraction, raw transactional records are aggregated into weekly time series suitable for forecasting.

The aggregation process groups observations by:

- deposit center;
- destination country;
- time index (weekly frequency).

Additional transformations are applied to compute:

- total shipment counts;
- aggregated weight metrics;
- descriptive statistics (mean, standard deviation).

The aggregation logic is illustrated in Figure 3.3.

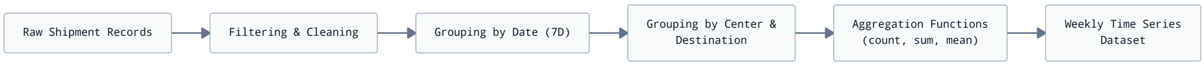

Figure 3.3 - Aggregation process used to construct weekly time series.

Figure 3.3 shows how transactional postal records are transformed into structured weekly time series. This step is critical because forecasting models require consistent temporal spacing and homogeneous series definitions.

3.2.3 Hierarchical Dataset Construction

The dataset is structured as a hierarchical time series, where multiple aggregation levels coexist within a unified representation.

The hierarchy includes:

- total outgoing international mail flow;
- deposit centers;
- destination countries;
- deposit center × destination combinations.

The hierarchical structure is constructed as shown in Figure 3.4.

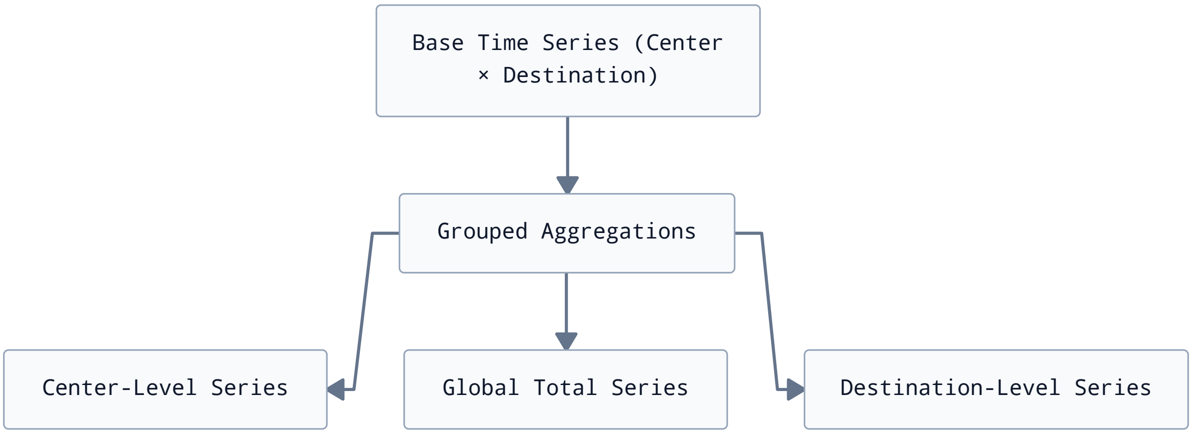

Figure 3.4 - Construction of hierarchical time series structure.

Figure 3.4 illustrates how multiple aggregation levels are derived from a single base dataset. This structure enables later application of hierarchical forecasting and reconciliation methods.

3.2.4 Data Quality and Storage Layer

To ensure reproducibility and efficiency, intermediate datasets are stored in compressed columnar formats (Parquet).

This choice is motivated by:

- reduced storage size;
- faster read/write performance;
- compatibility with Polars and Pandas workflows.

The data pipeline also includes validation steps to ensure consistency, including:

- removal of null or corrupted records;
- verification of temporal continuity;
- enforcement of unique identifiers for hierarchical series.

The data engineering layer provides the foundation of the forecasting system. It ensures that raw operational data is transformed into a structured, consistent, and hierarchical dataset suitable for modeling.

Without this layer, forecasting models would not be able to operate reliably due to inconsistencies in source systems and the absence of analytical data infrastructure.

The next section presents the implementation of feature engineering procedures applied on top of this dataset.

3.3 Feature Engineering Implementation

This section describes the implementation of feature engineering procedures applied to the structured dataset produced in Section 3.2. The objective of this layer is to enrich the raw time-series information with explanatory variables that improve forecasting performance across statistical, machine learning, and neural models.

The feature engineering design follows the structure defined in Section 2.4, with a separation between shared features, machine learning-specific features, and transformations applied uniformly across all models.

3.3.1 Shared Features (All Models)

A common set of features is used across all forecasting approaches to ensure comparability between model families.

The main shared feature is the deterministic trend component, constructed from the time index.

This feature is generated using a preprocessing pipeline and is included in both machine learning and neural forecasting models.

In addition, all models operate on a standardized weekly time index (ds) and a target variable representing shipment counts or aggregated weights.

The role of shared features in the modeling pipeline is summarized in Figure 3.5.

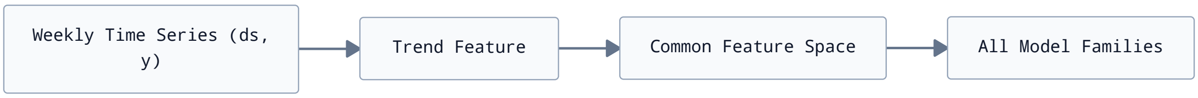

Figure 3.5 - Shared feature structure used across all forecasting models.

Figure 3.5 illustrates that all forecasting models operate on a unified feature representation. This ensures that performance differences between models are attributable to their learning mechanisms rather than differences in input representation.

3.3.2 Machine Learning Feature Set (MLForecast)

Machine learning models reformulate the forecasting problem as supervised learning. As a result, additional lag-based features are required to capture temporal dependencies.

The feature set used for ML models includes:

- lag features (1, 2, 3, 4, seasonal lag);
- deterministic trend feature;
- optional calendar-derived features (not retained in the final configuration).

The lag structure is defined as:

- y_{t-1}, y_{t-2}, y_{t-3}, y_{t-4};
- y_{t-m} where m corresponds to the seasonal period (approximately one Hijri year in weeks).

The ML feature generation process is illustrated in Figure 3.6.

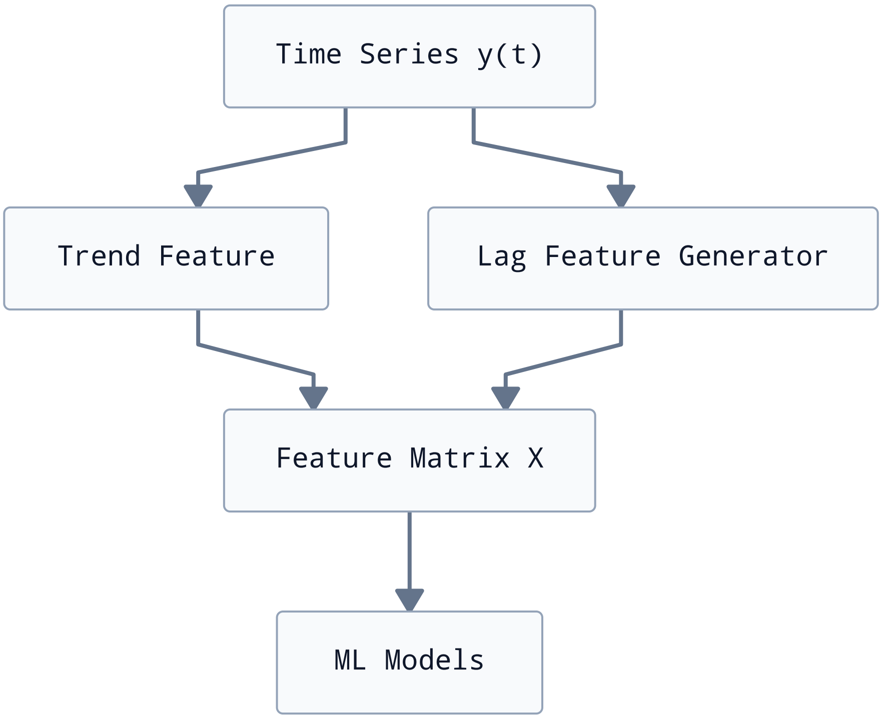

Figure 3.6 - Feature engineering pipeline for machine learning models.

Figure 3.6 shows how lagged observations and trend information are combined into a supervised learning dataset. This transformation is essential for enabling regression-based forecasting models.

3.3.3 Neural Feature Set (NeuralForecast)

Neural forecasting models differ from classical machine learning approaches in that they learn temporal representations directly from sequences of observations.

In this implementation, neural models are provided with:

- historical time windows (input sequences);
- optional trend feature shared with other models;
- no explicit lag feature engineering.

This design is consistent with the requirements of architectures such as NHITS and BiTCN, which internally learn temporal dependencies using convolutional or hierarchical structures.

The trend feature is retained to provide a global signal that improves stability in long-term pattern learning.

The neural input structure is summarized in Figure 3.7.

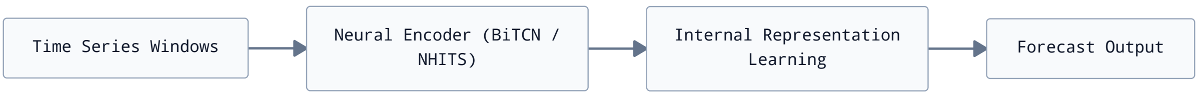

Figure 3.7 - Neural forecasting input-output structure.

Figure 3.7 illustrates that neural models bypass explicit lag construction and instead rely on learned representations of sequential input windows.

3.3.4 Time-Series Transformations

To improve model stability, logarithmic transformations are applied to the target variable during training.

The transformation used is:

$$y' = \log(1 + y)$$

This transformation reduces the impact of extreme values and stabilizes variance in highly skewed distributions.

After forecasting, an inverse transformation is applied:

$$y = \exp(y') - 1$$

The feature engineering layer provides a unified representation of the time series adapted to different modeling paradigms.

By separating shared features, machine learning-specific lag structures, and neural input representations, the system ensures consistency across models while preserving methodological flexibility.

The next section presents the implementation of the forecasting models themselves across statistical, machine learning, and deep learning frameworks.

3.4 Model Implementation

This section presents the implementation of the forecasting models used in this internship. The system follows a multi-paradigm strategy combining statistical, machine learning, and neural forecasting approaches.

All models are trained on the same hierarchical dataset and evaluated under a unified cross-validation framework to ensure comparability. The implementation is designed to reflect the modeling strategy defined in Section 2.5.

3.4.1 Statistical Models

Statistical models are used as interpretable baselines and as robust references for comparison against more complex approaches. They rely on explicit assumptions about temporal structure such as seasonality, trend, and autocorrelation.

The statistical modeling family used in this study is summarized in Figure 3.8.

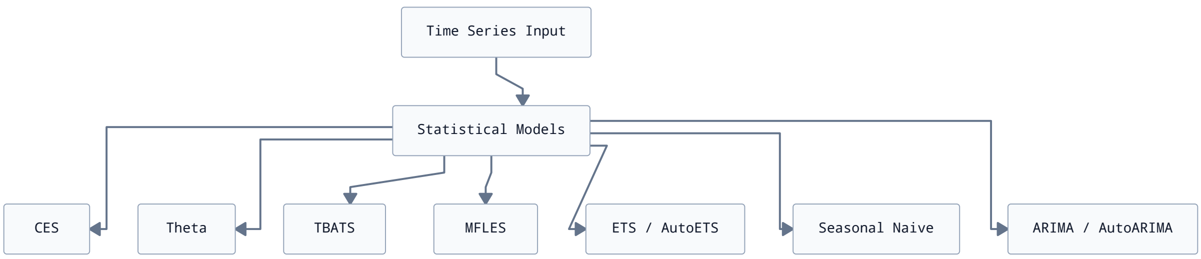

Figure 3.8 - Statistical forecasting model family used in StatsForecast.

Figure 3.8 shows the diversity of statistical models used in the system, ranging from simple seasonal baselines to more advanced decomposition-based approaches. These models serve as strong benchmarks for evaluating the added value of machine learning and neural methods.

The implementation is based on the StatsForecast framework:

```python
sf = StatsForecast(
  freq = "1w",
  n_jobs = -1,
  models = [
    SeasonalNaive(season_length = hijri_year_weeks),
    AutoETS(model = "AAA", damped = true),
    AutoARIMA(season_length = hijri_year_weeks),
    AutoCES(season_length = hijri_year_weeks),
    AutoMFLES(season_length = hijri_year_weeks),
    AutoTBATS(season_length = hijri_year_weeks),
    AutoTheta(season_length = hijri_year_weeks)
  ]
)
```

The models are trained on the full historical dataset and used both for forecasting and cross-validation evaluation.

The inclusion of multiple statistical models is motivated by their strong performance on structured seasonal data. In particular, models such as ETS and seasonal naive often remain competitive in short-horizon forecasting tasks, especially when data is noisy or limited.

In the observed results (Section 3.6), statistical models typically provide a stable baseline but are gradually outperformed by machine learning and neural approaches as more historical data becomes available and feature richness increases. This behavior is particularly visible when comparing cross-validation scores across cutoff dates.

3.4.2 Machine Learning Models

Machine learning models reformulate forecasting as a supervised regression problem using lagged features and engineered predictors. These models are implemented using the MLForecast framework.

The machine learning model family is summarized in Figure 3.9.

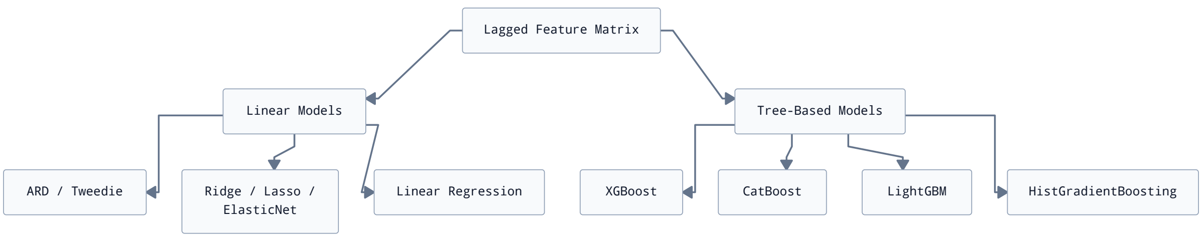

Figure 3.9 - Machine learning forecasting models implemented via MLForecast.

Figure 3.9 illustrates the separation between linear and non-linear model families. Linear models provide interpretability and stability, while gradient-boosted trees capture nonlinear interactions between lag features and seasonal patterns.

The implemented configuration is:

python

```python
fcst = MLForecast(
  models = [
    LinearRegression(),
    ElasticNet(),
    ARDRegression(),
    TweedieRegressor(),
    Lasso(),
    Ridge(),
    HistGradientBoostingRegressor(),
    LGBMRegressor(random_state = 1),
    XGBRegressor(random_state = 1),
    CatBoostRegressor(random_state = 1)
  ],
  freq = "weekly",
  lags = [1, 2, 3, 4, seasonality]
)
```

All models share the same feature set, including lag variables and a deterministic trend component. This ensures that performance differences reflect model capacity rather than feature availability.

3.4.3 Neural Models

Neural forecasting models are used to capture nonlinear temporal dependencies directly from sequential input windows. In the final implementation of this internship, only two architectures are retained: BiTCN and NHITS, both implemented using the NeuralForecast framework.

This choice was motivated by empirical stability during training and consistent performance under cross-validation compared to alternative neural architectures (e.g., LSTM or generic MLP), which were evaluated but not retained in the final pipeline. The retained neural architectures are summarized in Figure 3.10.

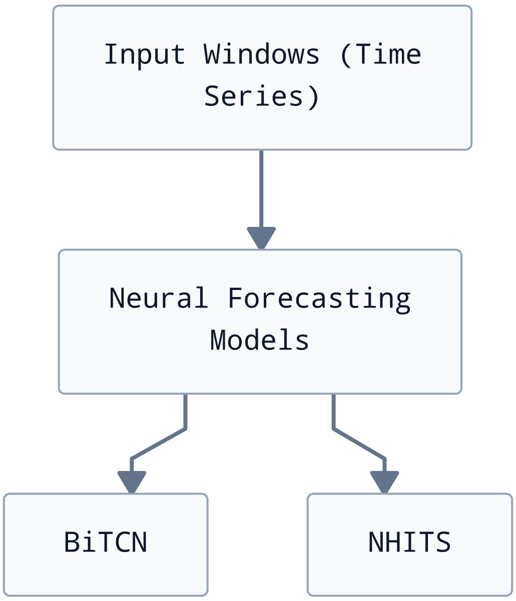

Figure 3.10 - Final neural forecasting models used in the internship (BiTCN and NHITS).

Figure 3.10 shows that the final neural layer of the system is intentionally reduced to two architectures. This simplification improves interpretability of results and reduces variance introduced by unstable training runs.

The final neural configuration is:

```python
nf = NeuralForecast(
  models = [
    AutoBiTCN(h = 4),
    AutoNHITS(h = 4),
  ],
  freq = "W"
)
```

Both models operate on the same preprocessed time series representation and share the same forecasting horizon and evaluation protocol as all other model families. In practice, the restricted neural configuration provides a better trade-off between complexity and robustness.

BiTCN captures local temporal dependencies through convolutional filters, while NHITS provides a hierarchical decomposition of the signal across multiple temporal resolutions. This complementarity was sufficient to cover the nonlinear patterns observed in international mail flows without introducing unnecessary model redundancy.

3.5 Hierarchical Forecasting Implementation

This section describes the implementation of the hierarchical forecasting layer used to ensure consistency between forecasts produced at different aggregation levels. The system follows a structured pipeline combining bottom-level prediction generation and top-down reconciliation.

3.5.1 Hierarchical Structure in the Data

The forecasting problem is defined over a hierarchical structure composed of multiple aggregation levels:

- total international outgoing mail flow;
- center-level series (Centre_Agence_depot);
- destination-level series (Destination);
- cross-level combinations (center × destination).

Each level represents a different granularity of the same underlying process, which introduces coherence constraints between forecasts.

This structure is explicitly encoded through composite identifiers of the form:

$$\text{unique\_id} = \text{Centre\_Agence\_depot} \times \text{Destination}$$

The hierarchical structure is summarized in Figure 3.11.

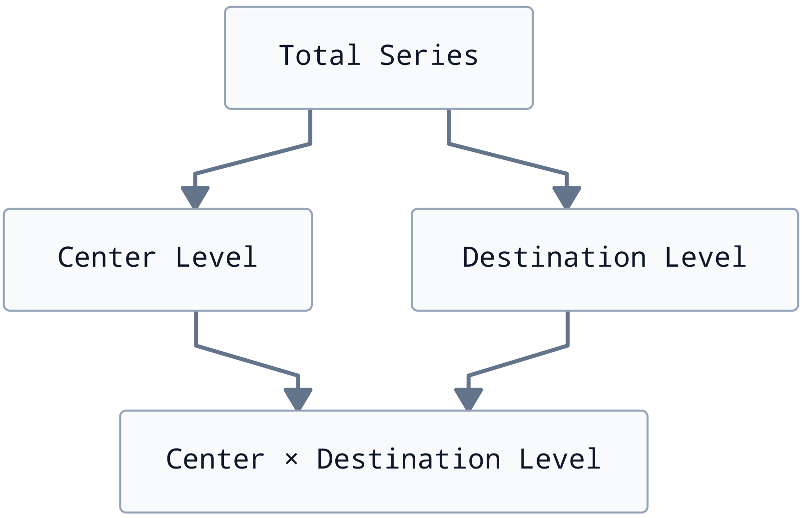

Figure 3.11 - Hierarchical structure of international mail time series.

Figure 3.11 illustrates the aggregation relationships between different levels of the forecasting hierarchy. The bottom level contains the most granular information, while higher levels represent aggregated views.

3.5.2 Bottom-Level Forecast Generation

All forecasting models (statistical, machine learning, and neural) are initially trained at the most granular level of the hierarchy.

Let:

$$\hat{y}_{i,t+h}$$

represent the forecast for series i at horizon h.

Each bottom-level forecast is later aggregated to higher levels before reconciliation.

3.5.3 Forecast Completion and Alignment

Before reconciliation, forecast outputs must be aligned with the hierarchical structure expected by the reconciliation procedure.

Some forecasting models only produce predictions for the aggregate international series, whereas reconciliation requires forecasts to be available for all nodes of the hierarchy and for every forecast horizon. Consequently, an intermediate completion step was implemented.

This procedure generates the full set of required identifier–timestamp combinations and aligns model outputs accordingly. Missing values are initialized to zero, since the subsequent Top-Down reconciliation step redistributes the aggregate forecasts using historical proportions.

The process is summarized in Figure 3.12.

As illustrated in Figure 3.12, the objective of this step is not to estimate additional forecasts, but simply to ensure that the forecast dataset conforms to the structure required by the reconciliation algorithm.

3.5.4 Top-Down Reconciliation Strategy

Forecasts are reconciled using a Top-Down proportional allocation strategy implemented through the HierarchicalForecast library.

The core idea is to distribute forecasts from the top level to lower levels based on historical proportions.

Formally:

$$\hat{y}_{i,t+h} = p_i \cdot \hat{y}_{\text{Total},t+h}$$

where:

- p_i is the historical proportion of series i;
- $\hat{y}_{\text{Total},t+h}$ is the forecast at the root level.

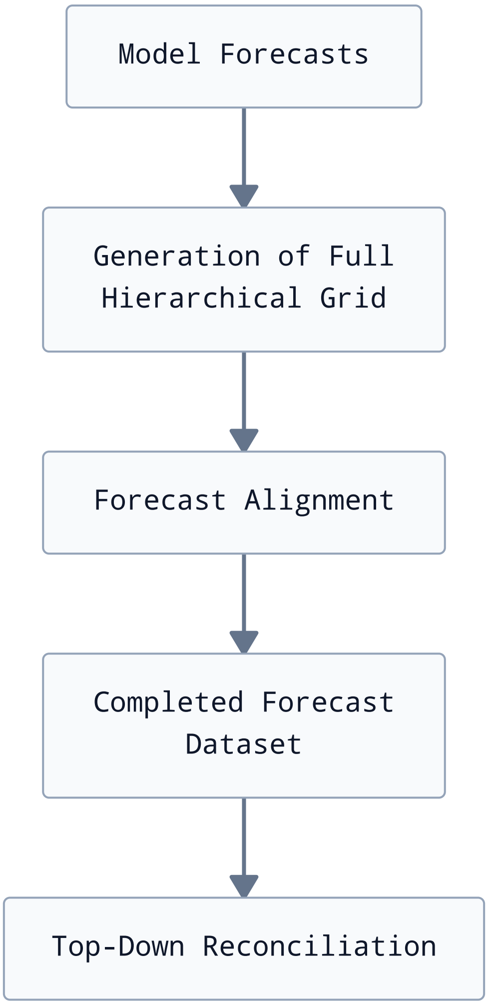

Figure 3.12 - Forecast completion and alignment performed before reconciliation.

The reconciliation process is illustrated in Figure 3.13.

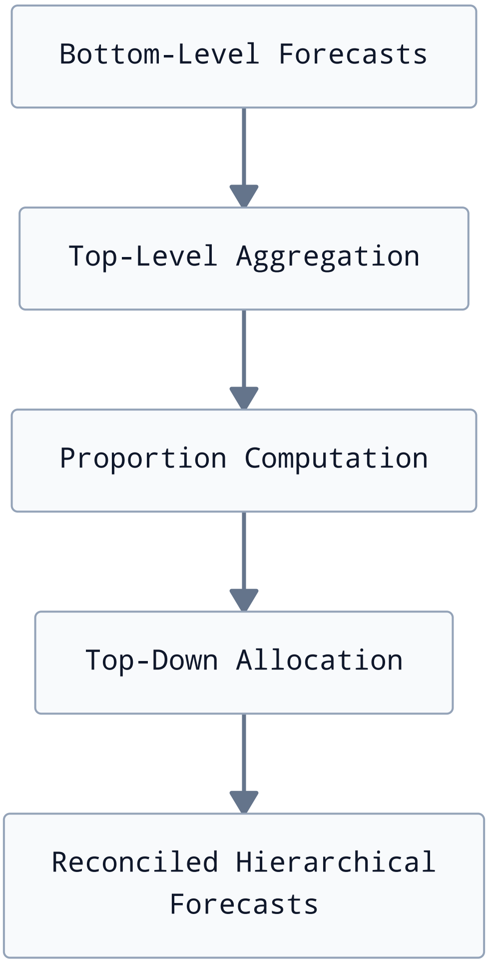

Figure 3.13 - Top-down hierarchical reconciliation process.

Figure 3.13 shows how bottom-level forecasts are adjusted to ensure coherence with higher aggregation levels. This step enforces structural consistency across the hierarchy.

3.5.5 Implementation Details

The reconciliation is implemented using the `HierarchicalForecast`[13] library with a Top-Down method based on historical average proportions[39].

Key steps include:

- construction of summing matrix S;
- definition of hierarchical tags for each series;
- application of reconciliation to model forecasts.

The implementation integrates directly into the pipeline:

```python
recon = HierarchicalReconciliation(
  reconcilers = [TopDown("proportion_averages")]
)
```

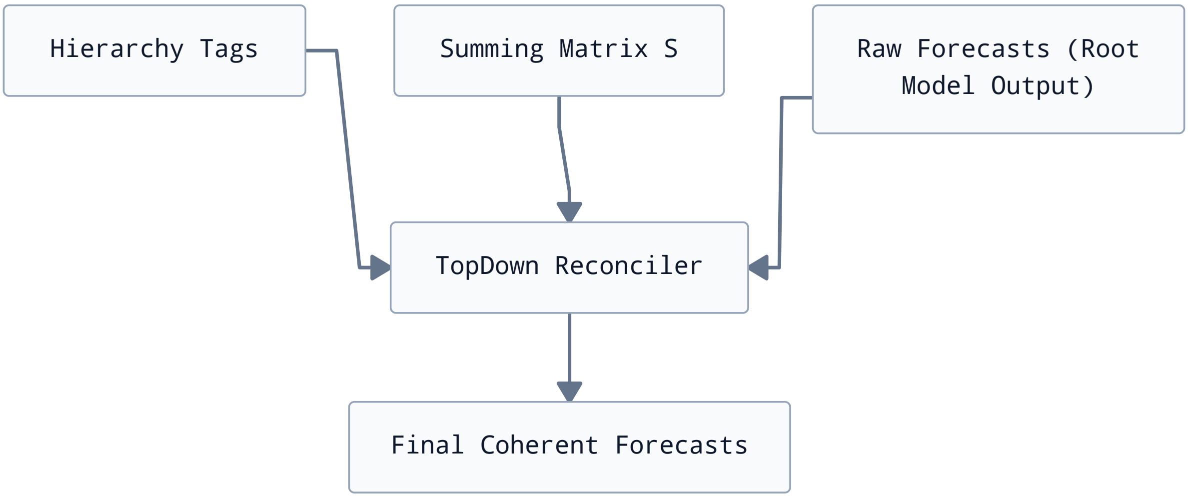

Figure 3.14 - Implementation of hierarchical reconciliation in the forecasting pipeline.

Figure 3.14 illustrates the integration of hierarchical metadata (tags and summing matrix) with model outputs to produce coherent forecasts.

The hierarchical forecasting layer ensures consistency between different aggregation levels of international mail flows. Without this step, independently generated forecasts would violate aggregation constraints and produce incoherent totals.

The combination of bottom-up generation and top-down reconciliation provides a practical compromise between model flexibility and structural correctness.

The next section presents the evaluation methodology used to assess forecasting performance across all model families.

3.6 Evaluation Results

This section presents the results obtained using the evaluation methodology described in Chapter 2. The objective is to compare the forecasting approaches considered in this study under identical experimental conditions and to identify models that provide the most suitable compromise between predictive accuracy, robustness, and operational applicability.

The comparison is based on rolling-origin cross-validation and relies on four complementary evaluation metrics: MASE, RMAE, ND, and SPIS. The analysis is organized by model family in order to facilitate interpretation and highlight the contribution of increasingly sophisticated forecasting techniques.

The presentation begins with statistical approaches, followed by machine learning models, neural architectures, and ensemble methods. Finally, all candidate models are compared jointly to support the selection of the forecasting approach retained for operational use.

3.6.1 Statistical Models Evaluation

Statistical forecasting models constitute an important baseline in time series analysis due to their relatively low computational requirements, ease of interpretation, and proven effectiveness in many industrial applications. In this study, seven statistical approaches were evaluated using the rolling-origin cross-validation framework introduced in Section 2.6.2.

Table 3.2 summarizes the average performances obtained by these models across all validation windows.

Table 3.2 - Cross-validation performances of statistical forecasting models.

Model	RMAE	MASE	ND	SPIS
AutoARIMA	0.462	643.479	0.147	70.211
CES	0.479	652.842	0.143	71.314
AutoTBATS	0.590	814.889	0.185	90.608
AutoTheta	0.698	948.811	0.182	102.857
AutoMFLES	0.706	907.191	0.201	97.903
AutoETS	0.933	1310.279	0.320	146.920
SeasonalNaive	1.000	1576.432	0.396	180.822

As shown in Table 3.2, AutoARIMA achieved the best overall performance among statistical approaches, obtaining an average RMAE of 0.462. CES exhibited very similar behavior, with only a marginal degradation in forecasting accuracy.

These results suggest that the studied series can be effectively modeled using approaches capable of capturing both autoregressive dynamics and seasonal patterns. The relatively good performance of CES also indicates that complex exponential smoothing mechanisms remain relevant for this forecasting problem.

AutoTBATS produced intermediate results despite its ability to model multiple seasonalities. This behavior may be explained by the limited length of the available historical dataset, which reduces the advantages associated with highly flexible decomposition-based methods.

AutoETS yielded the weakest performances among advanced statistical models and remained relatively close to the SeasonalNaive baseline. This observation suggests that the assumptions underlying traditional exponential smoothing are insufficient to adequately represent the variability observed in outgoing international mail flows.

Overall, statistical models provide robust and computationally efficient forecasting solutions. Although they are outperformed by more sophisticated approaches for short forecasting horizons, they remain attractive candidates for applications requiring stable long-term extrapolation.

3.6.2 Machine Learning Models Evaluation

Machine learning approaches were introduced to exploit nonlinear relationships and interactions between lagged observations and engineered temporal features. All models were trained using the same feature set described in Section 2.4.3 and evaluated using the cross-validation strategy presented in Section 2.6.2.

Table 3.3 presents the average performances obtained by machine learning models.

Table 3.3 - Cross-validation performances of machine learning models.

Model	RMAE	MASE	ND	SPIS
Ridge	0.404	576.802	0.134	64.322
LinearRegression	0.415	600.037	0.139	67.025
CatBoostRegressor	0.436	597.800	0.133	65.581
ARDRegression	0.484	756.803	0.182	86.264
XGBRegressor	0.513	693.310	0.144	75.522
HistGradientBoostingRegressor	0.594	792.219	0.174	85.559
TweedieRegressor	0.644	813.827	0.177	88.201
ElasticNet	0.685	844.689	0.190	91.930
LGBMRegressor	0.596	879.548	0.195	96.972
Lasso	0.746	899.170	0.205	97.537

According to Table 3.3, Ridge regression achieved the best overall performance among machine learning approaches, with an average RMAE of 0.404. LinearRegression exhibited comparable behavior, suggesting that a significant proportion of the predictive signal can be captured through linear relationships between lagged observations and temporal features.

CatBoostRegressor produced competitive results and achieved the lowest ND value within this family, indicating good point forecasting capabilities. However, its advantage over simpler linear methods remained relatively limited.

Surprisingly, gradient boosting approaches such as XGBoost, HistGradientBoostingRegressor, and LightGBM did not outperform regularized linear models. This may be partially explained by the relatively small number of observations available for training and the strong seasonal structure already captured by engineered lag features.

Regularized models such as ElasticNet and Lasso obtained lower rankings, indicating that aggressive feature shrinkage may remove useful information from the forecasting problem.

Overall, machine learning approaches significantly improved upon purely statistical methods while maintaining moderate computational requirements. They therefore represent an attractive compromise between forecasting accuracy and implementation complexity.

3.6.3 Deep Learning Models Evaluation

Deep learning approaches were investigated to determine whether more expressive architectures could better capture the complex temporal dependencies observed in outgoing international mail flows. Unlike machine learning methods relying explicitly on manually engineered lagged features, neural forecasting models learn internal representations directly from the historical sequence.

As described in Section 2.5, two neural architectures were retained in this study: BiTCN and NHITS. Both models were trained on the transformed weekly aggregated series and received the same trend-based exogenous features used by the other forecasting approaches.

Table 3.4 summarizes the performances obtained by these models over all cross-validation windows.

Table 3.4 - Cross-validation performances of neural forecasting models.

Model	RMAE	MASE	ND	SPIS
BiTCN	0.108	228.094	0.115	33.487
NHITS	0.137	289.796	0.146	42.546

As shown in Table 3.4, neural approaches substantially outperformed all previously evaluated forecasting families. BiTCN achieved the best overall results among all candidate models, obtaining the lowest values for all evaluation metrics considered.

Compared to the SeasonalNaive baseline, BiTCN reduced the average forecasting error by almost 90%, with an RMAE value of only 0.108. NHITS also demonstrated excellent predictive capabilities, ranking among the best-performing models despite slightly higher forecasting errors.

The superior performances of BiTCN may be attributed to its temporal convolutional architecture, which is particularly well suited for learning local temporal dependencies while preserving long-range information through dilated convolutions. This capability appears beneficial for modeling the recurrent seasonal patterns and short-term fluctuations characterizing international mail volumes.

However, it should be emphasized that these results correspond exclusively to the operational forecasting horizon considered in this study (**h = 4 weeks**). Additional experiments performed on longer horizons revealed a significant deterioration in neural forecasting quality. Although deep learning models provide outstanding short-term predictions, their forecasts become increasingly unstable as the prediction horizon grows.

This behavior contrasts with some statistical approaches, particularly AutoARIMA and CES, which tend to maintain more stable long-term trajectories. Consequently, the adoption of BiTCN within this work is motivated by its suitability for short-term operational planning rather than its ability to generate reliable long-range forecasts.

3.6.4 Ensemble Models Evaluation

In addition to evaluating individual forecasting models, ensemble approaches were investigated in order to assess whether combining complementary predictors could improve forecasting robustness.

Three ensemble forecasts were constructed following the methodology described in Section 2.6.2. Each ensemble corresponds to the arithmetic mean of the predictions produced by the two best-performing models within a given methodological family.

Table 3.5 presents the performances obtained by these ensemble forecasts.

Table 3.5 - Cross-validation performances of ensemble forecasting models.

Model	RMAE	MASE	ND	SPIS
stat_ensemble	0.431	604.645	0.135	66.234
ml_ensemble	0.385	538.302	0.124	59.577
neural_ensemble	0.112	237.953	0.120	34.935

According to Table 3.5, ensemble forecasts consistently improved upon most individual models within their respective families.

The statistical ensemble combining AutoARIMA and CES outperformed each of its constituent models individually, suggesting that averaging forecasts helps reduce model-specific errors.

Similarly, the machine learning ensemble constructed from Ridge and CatBoostRegressor achieved better performances than either model considered separately.

The neural ensemble combining BiTCN and NHITS also produced highly competitive results. Nevertheless, its performances remained slightly below those of BiTCN alone. This observation indicates that NHITS introduces a small amount of additional forecasting variance that offsets the potential benefits of model averaging.

Overall, ensemble methods demonstrated their ability to increase forecast robustness. However, in the present study, they did not systematically surpass the best individual models.

3.6.5 Final Model Selection

The previous sections evaluated candidate forecasting approaches within their respective methodological families. In order to select a forecasting model suitable for operational deployment, all evaluated approaches were compared jointly using the average cross-validation metrics presented in Chapter 2.

Since RMAE directly quantifies forecasting performance relative to the SeasonalNaive baseline, it was adopted as the primary ranking criterion. Secondary metrics (MASE, ND, and SPIS) were used to confirm the consistency of the rankings obtained.

Table 3.6 presents the complete ranking of all evaluated forecasting approaches according to their average RMAE score.

Table 3.6 - Global ranking of forecasting models based on average RMAE.

Rank	Model	RMAE
1	BiTCN	0.108
2	neural_ensemble	0.112
3	NHITS	0.137
4	ml_ensemble	0.385
5	Ridge	0.404
6	LinearRegression	0.415
7	stat_ensemble	0.431
8	CatBoostRegressor	0.436
9	AutoARIMA	0.462

Rank	Model	RMAE
10	CES	0.479
11	ARDRegression	0.484
12	XGBRegressor	0.513
13	AutoTBATS	0.590
14	HistGradientBoostingRegressor	0.594
15	LGBMRegressor	0.596
16	TweedieRegressor	0.644
17	ElasticNet	0.685
18	AutoTheta	0.698
19	AutoMFLES	0.706
20	Lasso	0.746
21	AutoETS	0.933
22	SeasonalNaive	1.000

As shown in Table 3.6, neural forecasting approaches clearly dominate the ranking. BiTCN achieved the best overall performance, closely followed by the neural ensemble and NHITS.

The results also indicate that relatively simple machine learning methods, particularly Ridge regression and ordinary linear regression, remain highly competitive despite their lower modeling complexity. Their performances exceed those of all standalone statistical models.

Among statistical approaches, AutoARIMA and CES constitute the most effective alternatives, obtaining results comparable to those of the best machine learning models while maintaining a lower computational cost and a high degree of interpretability.

The ranking further highlights the limited contribution of highly regularized linear models and some decomposition-based statistical methods for the considered forecasting task.

The average cross-validation metrics presented previously summarize model performances over all validation windows. However, they do not provide information about the evolution of forecasting accuracy as additional historical observations become available.

To analyze this behavior, the forecasting errors obtained for each validation window were visualized as a function of the cutoff date.

Figure 3.15 illustrates the evolution of RMAE values over successive cross-validation windows for the main forecasting approaches.

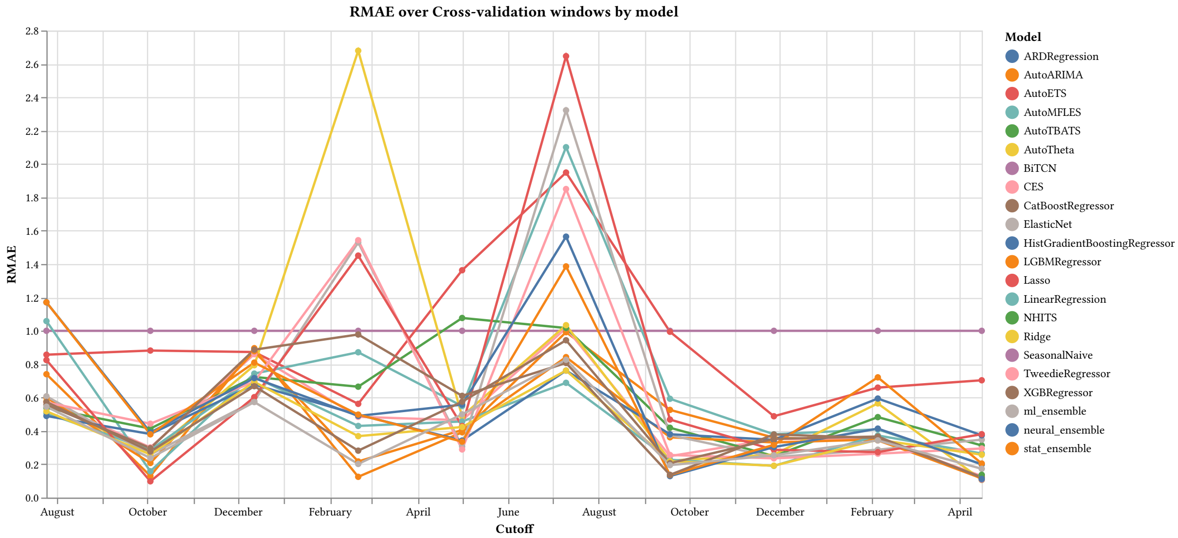

Figure 3.15 - Evolution of RMAE values across rolling cross-validation windows.

The behavior observed in Figure 3.15 suggests that forecasting performance depends not only on the choice of model but also on the amount of historical information available during training.

During the earliest validation windows, simpler approaches often remain competitive due to the relatively limited quantity of observations available for model estimation. As the training period expands, more expressive architectures progressively improve their predictive capabilities.

This phenomenon is particularly noticeable for BiTCN, whose forecasting errors decrease steadily over successive validation windows. The model appears to benefit substantially from larger training datasets, eventually becoming the most accurate forecasting approach considered in this study.

This observation is consistent with the greater representational capacity of neural architectures, which generally require a sufficient amount of historical data to fully exploit their modeling capabilities.

From an operational perspective, the objective of this study is to support short-term planning activities within Barid Al-Maghrib. The selected forecasting horizon corresponds to four weeks, which aligns with transport planning and resource allocation cycles.

For this specific use case, BiTCN constitutes the most appropriate forecasting model. It consistently achieved the best performances across all evaluation metrics and demonstrated increasing predictive capabilities as additional historical observations became available.

Nevertheless, complementary experiments conducted using longer forecasting horizons revealed a significant degradation in the quality of neural forecasts. Although BiTCN and NHITS provide outstanding short-term predictions, their forecasts tend to diverge more rapidly when extrapolated over extended periods.

Conversely, statistical approaches such as AutoARIMA and CES exhibit comparatively more stable long-term behavior. Consequently, the choice of BiTCN should not be interpreted as a universally superior forecasting solution, but rather as the model best aligned with the operational requirements of Barid Al-Maghrib, where decision-making primarily relies on short-term forecasts.

Based on these considerations, BiTCN was identified as the most suitable model for the primary operational use case considered in this study, namely four-week forecasting.

However, the forecasting system was designed to remain model-agnostic. Forecasts generated by all evaluated models are preserved, reconciled, and made available through the interactive visualization interface, allowing users to compare alternative forecasting approaches according to their specific analytical needs.

3.7 Forecast Outputs

This section presents examples of forecasts generated by the forecasting pipeline. The objective is not to compare forecasting models, as this was addressed in Section 3.6, but rather to illustrate the types of outputs produced by the system and their suitability for operational use.

Two forecasting horizons were considered. The first corresponds to the four-week planning horizon retained for operational decision-making, while the second illustrates the behavior of models over a significantly longer extrapolation period.

3.7.1 Short-term Forecasts (h = 4)

The primary objective of this study is to support short-term planning activities within Barid Al-Maghrib. Consequently, the forecasting horizon adopted throughout the evaluation process corresponds to four weeks ahead.

Forecasts generated over this horizon can support several operational activities, including:

- Allocation of transportation capacities;
- Anticipation of sorting center workloads;
- Human resource planning;
- Monitoring expected fluctuations in international shipment volumes.

Figure 3.16 presents an example of a four-week forecast generated by the system.

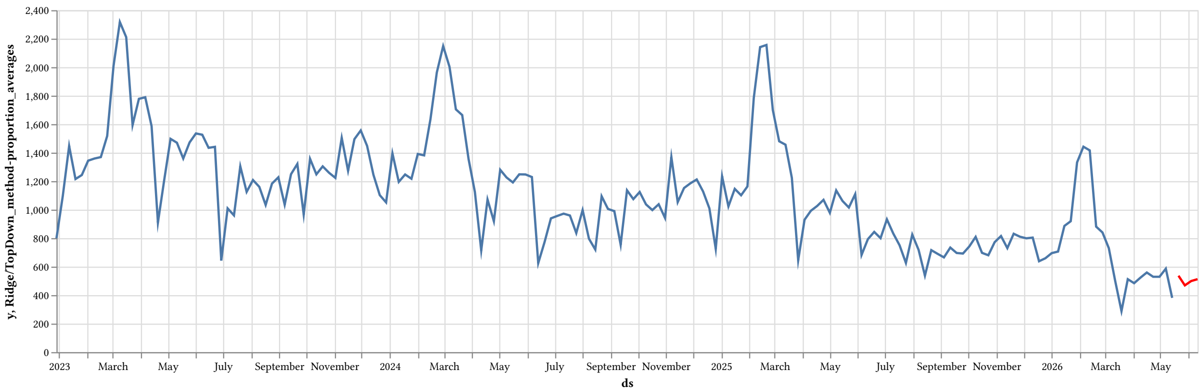

Figure 3.16 - Example of a four-week forecast of package count produced by the forecasting pipeline.

As shown in Figure 3.16, the forecasting system extends the historical trajectory while preserving the main seasonal characteristics observed in the training data. The relatively short prediction horizon limits uncertainty accumulation and allows complex models such as BiTCN to exploit local temporal patterns effectively.

For operational use, this forecasting horizon represents the most relevant scenario, since planning decisions at Barid Al-Maghrib are generally performed over relatively short time intervals.

3.7.2 Extended Forecasts (h = 100)

Although the forecasting problem addressed in this internship focuses primarily on short-term planning, additional experiments were conducted using an extended forecasting horizon of one hundred weeks.

The objective of these experiments was to analyze the stability of different forecasting approaches when producing long-range extrapolations.

Figure 3.17 illustrates an example of forecasts generated over an extended horizon.

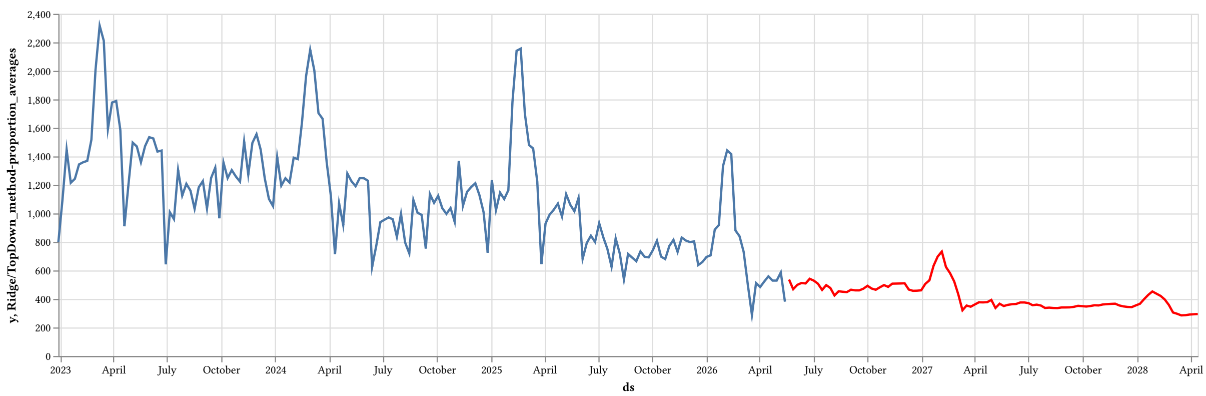

Figure 3.17 - Example of forecasts of package count generated over a horizon of one hundred weeks.

The results indicate significant differences between forecasting families.

Neural forecasting models, particularly BiTCN and NHITS, demonstrated excellent performances for short-term predictions but tended to exhibit increasing instability as the forecasting horizon expanded. Small deviations introduced during the first predicted periods progressively accumulated, leading to trajectories that became less realistic over time.

Conversely, statistical models such as AutoARIMA and CES generally produced smoother and more stable long-term forecasts. Although their short-term accuracy was lower than that of neural approaches, their extrapolation behavior appeared more robust for horizons substantially exceeding the operational requirements considered in this study.

These observations reinforce the importance of aligning model selection with the intended forecasting horizon. Since the objective of this internship is to support four-week operational planning cycles, the superior short-term performances of BiTCN justify its recommendation despite its weaker long-range behavior.

3.7.3 Forecast Coherence Across Hierarchical Levels

One advantage of the proposed approach lies in the ability to generate coherent forecasts simultaneously at multiple aggregation levels.

Examples include:

- total outgoing international mail;
- individual deposit centers;
- destination countries;
- combinations of deposit centers and destinations.

The reconciliation process guarantees that lower-level predictions remain compatible with aggregated forecasts.

The forecast associated with the deposit center located in Rabat is shown in Figure 3.18.

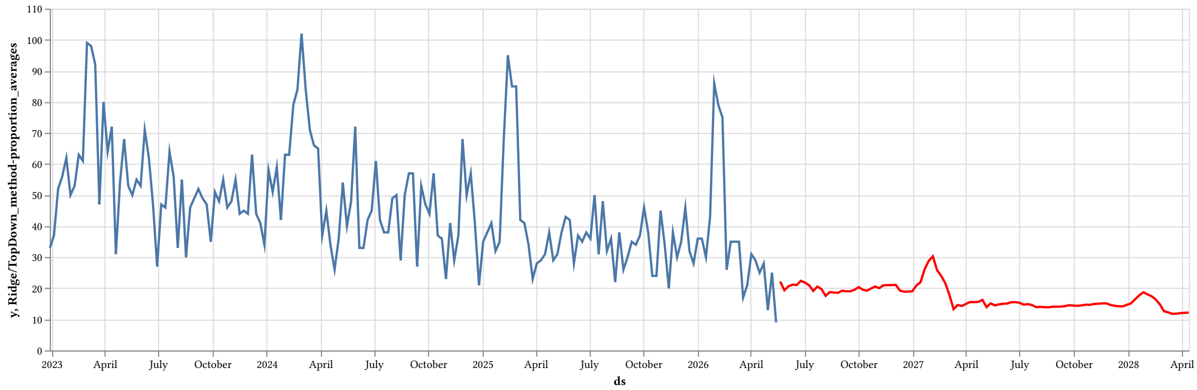

Figure 3.18 - Forecast for the Rabat deposit center after hierarchical reconciliation.

Figure 3.18 illustrates the temporal evolution of outgoing shipments originating from Rabat. The reconciled forecast remains compatible with the total forecast while preserving local dynamics specific to this center.

The forecast corresponding to shipments destined for the United States is presented in Figure 3.19.

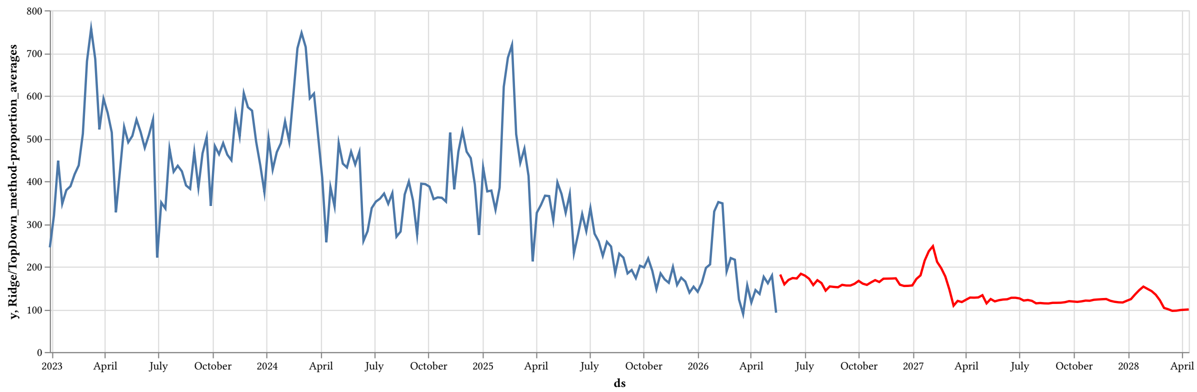

Figure 3.19 - Forecast for shipments destined to the United States.

Figure 3.19 highlights the demand dynamics associated with one of the major international destinations observed in the dataset. Such forecasts can support transport planning and resource allocation decisions for destination-specific flows.

Finally, a more granular example involving shipments deposited in Rabat and destined for the United States is shown in Figure 3.20.

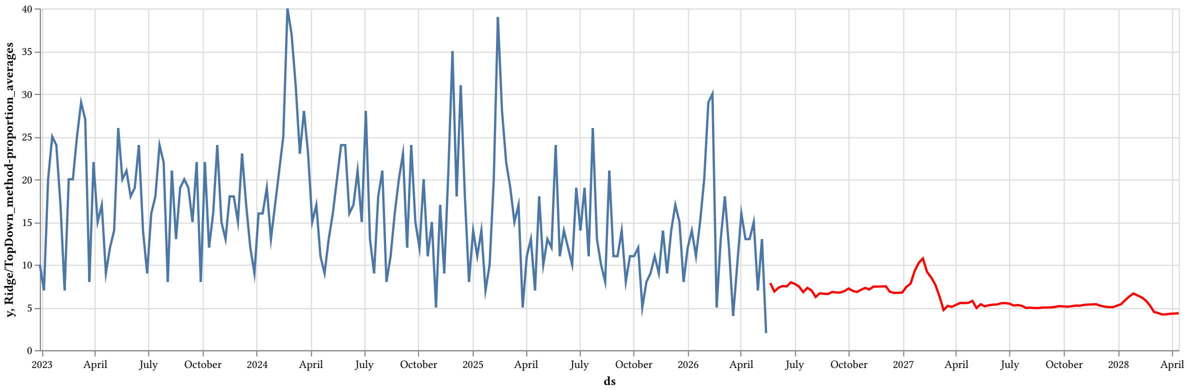

Figure 3.20 - Forecast for shipments from Rabat to the United States.

Figure 3.20 demonstrates the ability of the proposed system to operate at the lowest level of the hierarchy. Although such series are generally noisier and contain fewer observations, reconciliation guarantees that their forecasts remain coherent with higher aggregation levels.

These examples illustrate the flexibility of the forecasting framework. Users can explore predictions at the national level, for individual deposit centers, for destination countries, or for specific origin-destination pairs without violating aggregation constraints.

The availability of coherent forecasts at multiple levels is particularly valuable in operational settings, where planning decisions may be taken by regional managers, international logistics teams, or central management. To facilitate this exploration process, an interactive visualization interface was developed and is presented in the next section.

3.8 Interactive Visualization System

3.9 Interactive Visualization System

The forecasting pipeline developed during this internship does not terminate at the model inference stage. To facilitate the interpretation and exploration of forecasting results, an interactive visualization layer was implemented using Altair.

The objective of this component is to provide an intuitive interface for analyzing historical observations and reconciled forecasts at different levels of aggregation. Rather than restricting users to a single forecasting approach, the visualization system exposes the outputs generated by all evaluated models, allowing interactive comparisons and supporting exploratory analysis.

This visualization layer constitutes the final stage of the operationalization process and serves as a decision-support tool for understanding future international mail flow dynamics.

3.9.1 Altair Dashboard Architecture

The dashboard was developed using Altair[40], a declarative statistical visualization library based on the Vega-Lite[41] grammar.

Unlike traditional dashboards that perform model inference on demand, the proposed system relies on forecasts generated offline during the forecasting phase. Once forecasts have been produced, transformed back to their original scale, and reconciled hierarchically, they are stored within tabular structures suitable for visualization.

Figure 3.21 illustrates the logical organization of the visualization system.

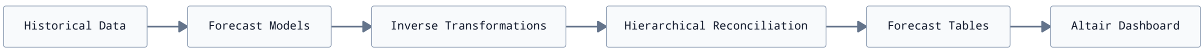

Figure 3.21 - Architecture of the interactive visualization system.

As shown in Figure 3.21, the dashboard consumes already computed forecasts. Consequently, user interactions do not trigger additional forecasting computations, resulting in fast response times and low computational requirements.

3.9.2 Visualization Components

The visualization system combines historical observations and forecasting outputs within the same graphical representation.

Historical observations are extracted from the aggregated time series used during model training, while forecasted values correspond to reconciled predictions obtained after hierarchical post-processing.

The dashboard is composed of two main visualization layers:

- A historical series represented as a line chart;
- A forecast series represented as an overlayed line chart.

These layers are combined into a single interactive visualization, enabling direct comparisons between observed and predicted trajectories.

Figure 3.22 presents an example of the resulting visualization.

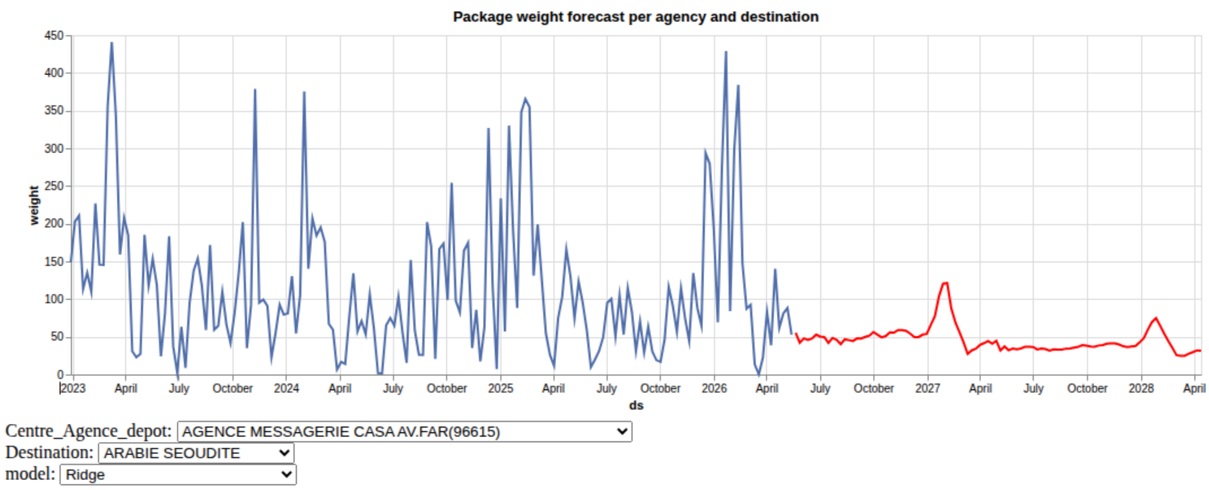

Figure 3.22 - Example of the interactive forecasting dashboard.

The superposition of observed and predicted values facilitates the assessment of trend continuity, seasonality preservation, and anticipated changes in future shipment volumes.

3.9.3 User Interaction Layer

The dashboard incorporates several interactive controls allowing users to explore forecasts dynamically.

Three independent selection mechanisms were implemented:

- Forecasting model selector;
- Origin agency selector;
- Destination selector.

These controls enable users to analyze forecasts at multiple hierarchical levels while simultaneously comparing the behaviors of different forecasting approaches.

Examples of possible analyses include:

- Total outgoing international flows;
- Forecasts for a specific deposit center;
- Forecasts associated with a particular destination country;
- Forecasts corresponding to a given origin–destination pair.

Selections are propagated through Altair parameter bindings and filtering mechanisms, allowing charts to update automatically without requiring additional server-side computations.

This approach provides a lightweight yet flexible exploration interface suitable for operational environments.

3.9.4 Operational Role

Beyond its visualization capabilities, the dashboard serves as a decision-support instrument capable of enhancing visibility over international mail flows.

Potential operational applications include:

- Anticipation of demand peaks;
- Capacity planning;
- Monitoring of specific international corridors;
- Comparison of alternative forecasting approaches;
- Exploration of hierarchical shipment patterns.

The possibility of interactively switching between forecasting models allows users to assess the robustness of predictions and identify situations where different modeling assumptions produce diverging trajectories.

Although the dashboard was developed primarily for analytical purposes within the scope of this internship, its architecture could be integrated into broader business intelligence initiatives and contribute to future data-driven decision-making processes within Barid Al-Maghrib.

3.10 Discussion

The results obtained during this internship demonstrate the feasibility of building a forecasting pipeline for outgoing international mail flows despite the limitations imposed by the operational information systems available at Barid Al-Maghrib.

From a data engineering perspective, a significant part of the effort was devoted to reconstructing an analytical dataset from heterogeneous operational sources. The absence of dedicated APIs, the fragmentation of historical information, and the constraints associated with legacy systems considerably increased the complexity of the data acquisition process. The implementation of automated extraction procedures and intermediate storage mechanisms nevertheless made it possible to obtain a consistent historical dataset suitable for time-series analysis.

The evaluation results indicate that neural forecasting models, and particularly BiTCN, constitute highly effective approaches for short-term forecasting in the considered context. Their ability to exploit local temporal dependencies enabled substantial improvements over traditional statistical baselines and machine learning methods.

However, the experiments also revealed several limitations. First, neural architectures exhibited reduced stability when forecasts were generated over long horizons. While these models achieved excellent performances for the four-week planning horizon adopted in this study, their predictive behavior became progressively less reliable as the forecasting horizon increased.

Second, the forecasting models considered in this work were developed exclusively from historical shipment observations and derived temporal features. Exogenous variables potentially influencing international mail volumes, such as public holidays, religious events, macroeconomic indicators, or major disruptions in international logistics networks, were not incorporated into the modeling process.

Finally, although the hierarchical reconciliation procedure guarantees coherence between aggregation levels, the adopted Top-Down approach remains relatively simple compared to more sophisticated reconciliation methods available in the literature, such as MinT-based techniques. Additional experiments could therefore be conducted to evaluate the impact of alternative reconciliation strategies.

Despite these limitations, the proposed methodology provides a practical framework for transforming operational postal data into actionable forecasting outputs. It demonstrates how recent forecasting libraries, modern data engineering practices, and interactive visualization tools can be combined to support decision-making activities in environments where analytical infrastructures remain limited.

Conclusion

This chapter presented the implementation and evaluation of the forecasting system developed during the internship.

The chapter described the main components of the proposed solution, including the data engineering pipeline, feature generation procedures, forecasting models, hierarchical reconciliation mechanisms, and interactive visualization layer. The experimental evaluation highlighted the superior short-term performances of neural approaches, particularly BiTCN, while also emphasizing the importance of considering forecasting horizons when selecting operational models.

The resulting system provides a complete workflow capable of extracting, processing, forecasting, reconciling, and visualizing international mail flows. Although developed within the scope of an internship, the proposed approach constitutes a first step toward the integration of predictive analytics capabilities into operational postal environments.

General Conclusion

This internship was conducted within Barid Al-Maghrib in a context characterized by the coexistence of large volumes of operational data and information systems primarily designed for transaction processing rather than analytical exploitation. The study focused on outgoing international mail flows, a domain whose variability directly impacts transport planning, workload distribution, and resource allocation.

The work carried out throughout the internship addressed the complete forecasting process, from data acquisition to the visualization of forecasting results. Due to the absence of dedicated analytical infrastructures and standardized extraction interfaces, particular attention was devoted to the development of robust data engineering procedures capable of reconstructing a coherent historical dataset from heterogeneous operational sources.

A forecasting framework aligned with the CRISP-DM methodology was subsequently designed and implemented. Several families of forecasting approaches were investigated, including statistical models, machine learning algorithms, and deep learning architectures. Their performances were assessed through a rolling-origin cross-validation strategy using multiple complementary evaluation metrics.

The experimental results demonstrated that neural forecasting approaches achieved the best performances for the four-week forecasting horizon considered in this study. In particular, the BiTCN architecture consistently outperformed competing methods according to all evaluation metrics. At the same time, additional experiments highlighted the importance of adapting model selection to the forecasting horizon, as statistical models exhibited greater stability for long-term extrapolation tasks.

Beyond forecasting performance, the proposed solution also integrates hierarchical reconciliation techniques and an interactive visualization layer developed using Altair. This component enables users to explore reconciled forecasts dynamically, compare forecasting models, and analyze shipment flows at different aggregation levels. The resulting system therefore constitutes a practical decision-support tool that bridges the gap between operational information systems and predictive analytics.

From a personal perspective, this internship provided an opportunity to consolidate competencies acquired during the engineering curriculum while gaining practical experience in a real industrial environment. It involved addressing challenges related to data extraction, time-series modeling, software integration, and the communication of analytical results to operational stakeholders.

Several perspectives can be envisaged to extend the present work. Future developments may include the incorporation of external explanatory variables, experimentation with alternative reconciliation techniques, the implementation of probabilistic forecasting approaches, and the integration of the proposed pipeline into existing business intelligence ecosystems within Barid Al-Maghrib.

Overall, this internship illustrates how modern data engineering practices and forecasting methodologies can contribute to improving visibility over postal demand and support the ongoing digital transformation initiatives undertaken by Barid Al-Maghrib.

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Appendix A

Data Acquisition and Exploratory Dataset Characterization

Introduction

This appendix documents the data acquisition procedures, exploratory investigations, and dataset characterization activities conducted during the internship. While the main body of the report focuses on the forecasting methodology and its implementation, a significant amount of engineering effort was devoted to reconstructing analytical datasets from operational information systems that were not originally designed for large-scale data analysis.

Several challenges were encountered during this process, including the absence of public APIs, limitations of legacy web interfaces, restricted export capabilities, and the need to process large volumes of semi-structured HTML documents. To address these issues, dedicated extraction, transformation, and archival workflows were developed using web automation tools, HTML parsers, and columnar storage formats.

This appendix is organized into four parts. The first part presents datasets directly extracted from the SMI reporting interface. The second part describes the automated acquisition and archival strategy adopted to collect shipment tracking information. The third part documents the analytical parquet datasets generated through the ETL pipeline and summarizes their statistical properties. Finally, a short synthesis highlights the contribution of these activities to the forecasting system developed in this internship.

A.1 SMI Envois par Produit International (`smi_envoisbyproduitintern`)

The `smi_envoisbyproduitintern` dataset constituted the primary source of information used to reconstruct historical shipment deposits. In particular, it provided the **Date_depot** variable, which was subsequently employed to build weekly aggregated time series.

No programmatic interface was available to access this data. Furthermore, the underlying web application exhibited significant performance limitations and frequently failed when requesting periods longer than six months. Consequently, a fully automated extraction workflow was developed using the Playwright library. The extraction process iterated over successive time intervals from January 2023 to May 2026, automatically interacting with date pickers, dropdown menus and CSV export functionalities exposed by the web portal.

Duplicate observations were removed using the `codeenvoi_` identifier, which acts as a shipment-level unique key. After the deduplication step, the dataset contained 195,858 observations.

Table A.1 summarizes the cardinality of the main variables. The number of distinct values associated with `codeenvoi_` is very close to the total number of observations, indicating that only a limited number of duplicate records were present in the raw exports.

Table A.1 - Cardinality summary for the `smi_envoisbyproduitintern` dataset

Feature	Unique values
`codeenvoi_`	195844
`datedepot`	117202
`origine`	2

Feature	Unique values
`destination_`	184
`site_depot_`	800
`pouds_reel_`	13354
`longueur_`	187
`largeur_`	218
`hauteur_`	114
`poids_volume_`	4272

Table A.2 presents descriptive statistics for the numerical and temporal variables available in the extracted data. Several dimensional attributes exhibit substantial missingness, whereas the shipment weight variable remains complete. These observations motivated the later decision to focus primarily on shipment counts and aggregate weights for forecasting purposes.

Table A.2 - Descriptive statistics for the `smi_envoisbyproduitintern` dataset

Statistic	`datedepot`	`pouds_reel_`	`longueur_`	`largeur_`	`hauteur_`	`poids_volume_`
Null Count	0	0	62073	62216	74233	0
Mean	2024-06-25 19:10:14	5.54	38.64	26.38	16.97	0.95
Std	-	6.02	30.07	24.78	10.82	10.15
Min	2023-01-02 00:00:00	0.008	0.0	0.0	0.0	0.0
25%	2023-09-05 10:18:00	1.25	26.0	20.0	10.0	0.0
50% (Median)	2024-05-17 09:06:00	3.3	36.0	23.0	14.0	0.0
75%	2025-03-17 13:57:00	7.74	50.0	30.0	22.0	0.0
Max	2026-05-18 15:51:00	67.58	7055.0	3715.0	1200.0	3084.585

The `smi_envoisbyproduitintern` dataset played a central role in the data acquisition process. Besides providing an initial view of shipment activity, it constituted the source from which all shipment identifiers (`codeenvoi_`) were extracted.

These identifiers were subsequently used to automate requests to the shipment tracking webpage. Since no bulk export mechanism or public API was available, each shipment had to be queried individually. The resulting HTML responses were archived and later transformed into structured parquet datasets used throughout the remainder of this work.

Consequently, `smi_envoisbyproduitintern` should be considered as the entry point of the complete data acquisition pipeline rather than merely an exploratory dataset.

A.2 SMI Suivi Expédition (`smi_suiviexpedition`)

The `smi_suiviexpedition` dataset was extracted during the exploratory phase of the internship to investigate the feasibility of analyzing shipments transiting through specific operational centers, particularly Laayoune.

Although the dataset was ultimately not incorporated into the forecasting pipeline presented in this report, it is documented here for completeness and because it may support future studies related to regional traffic analysis, transit monitoring, or service performance assessment.

After extraction and cleaning, the dataset contained 244,147 observations.

Table A.3 summarizes the cardinality of the principal variables.

Table A.3 - Cardinality summary of the `smi_suiviexpedition` dataset

Feature	Unique values
`CAB`	244147
`CLIENT`	4830
`SERVICES_OPTIONNELS`	165
`SITE_EXPEDITEUR`	107
`RECEPTION`	2
`DERNIER_STATUT`	9
`DATE_DERNIER_STATUT`	111994
`SITE_DERNIER_STATUT`	8
`DESTINATION`	568

The dataset exhibits a relatively large number of destinations and customers, suggesting a broad geographical and operational coverage.

Table A.4 presents descriptive statistics for the available temporal variable.

Table A.4 - Descriptive statistics of the `smi_suiviexpedition` dataset

Statistic	`DATE_DERNIER_STATUT`
Null Count	0
Mean	2024-08-21 21:07
Std	—
Min	2022-01-05 00:00:00
25%	2023-10-25 12:16:00
50% (Median)	2024-09-03 09:20:54
75%	2025-07-03 15:38:41
Max	2026-04-28 14:42:43

As shown in Table A.4, the dataset covers a temporal span comparable to the other operational datasets extracted during the internship. However, since it was not directly linked to the forecasting objective defined in Chapter 2, it was excluded from the final analytical workflow.

Nevertheless, the dataset could constitute a useful resource for future investigations focused on shipment routing, regional traffic characterization, or the analysis of operational bottlenecks within the postal network.

A.3 Parcel Tracking Data Extraction and Archival Strategy

The shipment tracking system provides detailed operational information describing the progression of parcels through the postal network. Unlike the datasets discussed previously, this information is not exposed through downloadable reports or programmatic interfaces. Instead, it is only accessible through an interactive webpage where users manually enter a shipment identifier and obtain the corresponding tracking history as an HTML table.

As explained in A.1, shipment identifiers were extracted from the `smi_envoisbyproduitintern` dataset and used as inputs for an automated collection procedure. A Playwright-based scraper was developed to query the tracking webpage for each identifier individually and download the resulting HTML page.

Figure A.1 illustrates the overall acquisition and transformation workflow.

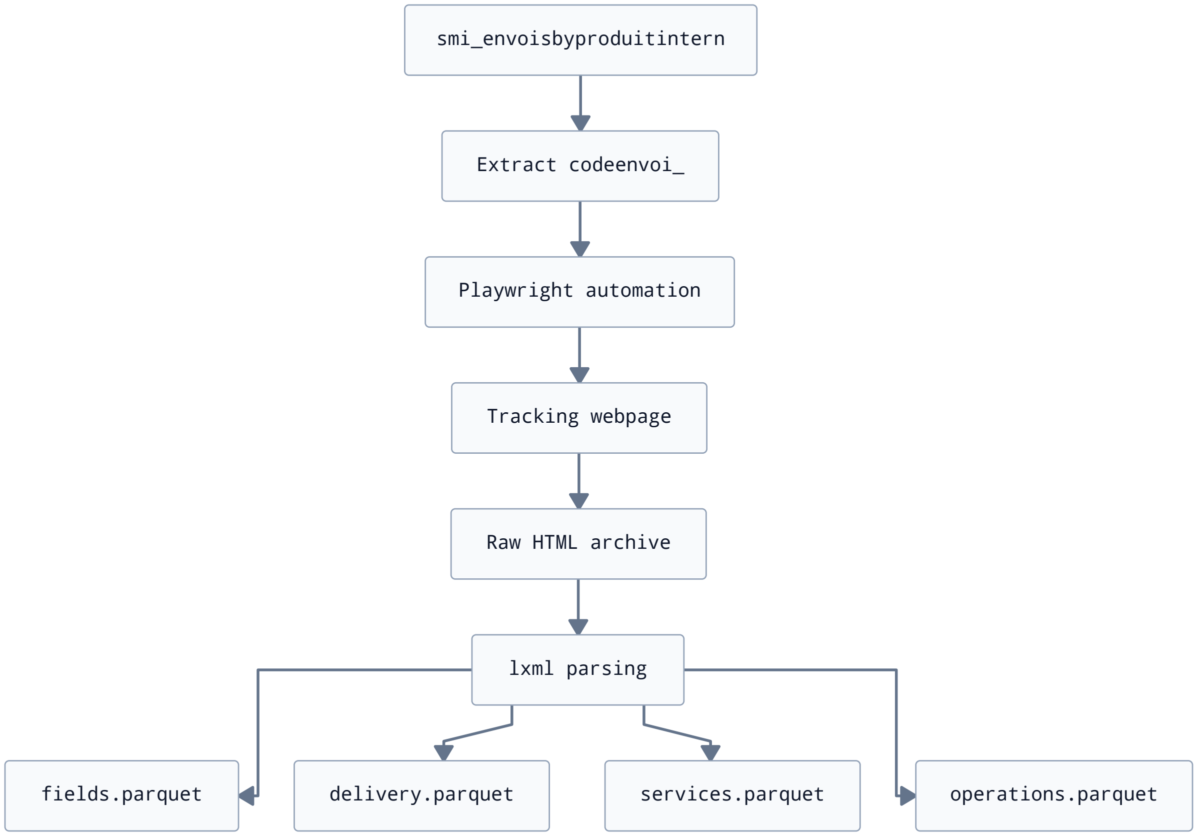

Figure A.1 - Tracking data acquisition pipeline

Figure A.1 highlights that the forecasting datasets were not directly available from the operational information systems. Instead, they had to be reconstructed through a dedicated extraction and transformation workflow specifically developed during the internship.

To preserve a complete source of truth, the raw HTML responses associated with approximately 234,000 shipments were archived locally before any processing step was performed. This archive initially occupied approximately 10.2 GB of storage.

An ETL procedure based on the `lxml` library was subsequently implemented to parse the downloaded HTML files and extract structured information. The extracted data were organized into four distinct parquet datasets:

- `fields.parquet`, containing shipment-level attributes and static information;
- `operations.parquet`, describing the sequence of operational events experienced by each shipment;
- `delivery.parquet`, containing delivery-related information and financial attributes;
- `services.parquet`, describing optional services associated with shipments.

The use of the Apache Parquet format considerably reduced storage requirements thanks to dictionary encoding and Snappy compression. The initial archive size of approximately 10.2 GB was reduced to roughly 35 MB, corresponding to a reduction of more than 99.6%.

Besides improving storage efficiency, the conversion to parquet substantially accelerated subsequent analytical operations, particularly filtering, aggregation, and exploratory queries performed using Polars.

The resulting parquet datasets formed the analytical foundation used throughout the remainder of this internship.

A.4 Analytical Datasets

The HTML parsing procedure described in A.3 produced four parquet datasets corresponding to different aspects of shipment information. While all datasets were explored during the internship, only the `fields` dataset was ultimately employed for the forecasting pipeline presented in Chapter 2 and Chapter 3.

The remaining datasets are nevertheless documented in this appendix because they may support future investigations related to shipment routing, delivery performance analysis, and customer service studies.

A.4.1 `fields.parquet`

The `fields` dataset contains shipment-level attributes extracted from the tracking pages. It combines operational characteristics, customer information, shipment dimensions, and destination data.

For the forecasting study, observations were filtered to retain only valid shipments. More specifically, records whose `Etat` variable indicated an invalid status were discarded. Additionally, only observations with a deposit date greater than or equal to January 1, 2023 were considered, as earlier periods exhibited substantial sparsity and would not provide a sufficiently stable basis for model training.

After preprocessing, the dataset contained 234,718 observations.

Table A.5 summarizes the cardinality of the principal variables contained in the dataset.

Table A.5 - Cardinality summary of the `fields` dataset

Feature	Unique values
`Cab`	234716
`Id`	234718
`Date_depot`	1041
`Type_cab`	4
`Dernier_statut`	10
`Regime`	3
`Contrat`	4843
`Etat`	1
`Poids_global_en_KG`	12435
`Centre_Agence_depot`	1484
`Destination`	563
`Client`	6892
`Produit_Niveau_Service`	16
`Mode_paiement`	7
`Taxe_DTQ_Dhs`	3
`Canal_de_livraison_1`	3
`Canal_de_livraison_2`	442
`Longueur`	296
`Hauteur`	114
`Largeur`	191
`Poids_Volumetrique`	39

The large number of distinct values associated with variables such as `Centre_Agence_depot`, `Destination`, and `Client` highlights the highly heterogeneous nature of international postal activity.

Table A.6 presents descriptive statistics for the principal numerical variables.

Table A.6 - Descriptive statistics of the `fields` dataset

Statistic	`Date_depot`	`Poids_global_en_KG`	`Taxe_DTQ_Dhs`	`Longueur`	`Hauteur`	`Largeur`	`Poids_Volumetrique`
Null Count	0	0	44484	202002	203394	202933	47777
Mean	2024-09-12	3.71	0.025	33.53	14.41	26.69	0.0003

Statistic	Date_depot	Poids_global_en_KG	Taxe_DTQ_Dhs	Longueur	Hauteur	Largeur	Poids_Volumetrique
Std	—	7.26	0.157	47.30	35.24	56.57	0.0424
Min	2023-01-02	0.0	0.0	0.0	0.0	0.0	0.0
25%	2023-11-22	0.12	0.0	20.0	7.0	14.0	0.0
50% (Median)	2024-09-24	0.76	0.0	25.0	10.0	20.0	0.0
75%	2025-07-08	4.04	0.0	30.0	12.0	20.0	0.0
Max	2026-04-28	1757.0	1.0	913.0	665.0	750.0	15.125

Several observations can be made from Table A.6. First, shipment weights exhibit considerable variability, ranging from very small packets to consignments exceeding one metric ton. Second, volumetric measurements are frequently unavailable, suggesting that these variables may not be systematically recorded during operational processing. Finally, the temporal distribution of deposit dates confirms that the selected study period provides a sufficiently dense and continuous basis for time-series modeling.

The `fields` dataset constitutes the primary analytical source employed throughout the forecasting workflow developed in this internship.

A.4.2 `operations.parquet`

The `operations` dataset contains the sequence of operational events recorded throughout the life cycle of each shipment. In contrast to the `fields` dataset, which provides a static description of parcels, this dataset captures the temporal evolution of their processing within the postal network.

The original `Etat` field contained composite textual information describing both the operation type and its validation status. To facilitate subsequent analyses, a preprocessing step based on regular expressions was implemented to extract two normalized variables:

- `status_code`, representing the operation category;
- `is_valid`, indicating whether the operation was validated by the information system.

For example, the value

```text
[ depot ] VALIDE (V)
```

was transformed into

```text
status_code = "depot"
is_valid = "V"
```

Only validated operations were retained for analysis. Furthermore, a filter based on `status_code` was applied to restrict the dataset to outgoing shipment events, as incoming international shipments represented only a negligible fraction of the available observations.

After preprocessing, the dataset contained 2,077,161 records.

Table A.7 summarizes the cardinality of the principal variables contained in the dataset.

Table A.7 - Cardinality summary of the `operations` dataset

Feature	Unique values
`cab`	244091
`id`	257828
`Date_operation`	1546
`Heure_Syst_Oper`	438442
`Statut`	29
`Agence`	1574
`Agent_oper`	5475
`Etat`	14
`Date_Etat`	1359
`Agent_maj`	2
`ORIGINE`	5
`status_code`	14
`is_valid`	1

The large number of operational records relative to the number of shipments reflects the fact that parcels generally undergo multiple processing stages before reaching their final destination.

Table A.8 presents descriptive statistics for the temporal variables available in the dataset.

Table A.8 - Descriptive statistics of the `operations` dataset

Statistic	`Date_operation`	`Heure_Syst_Oper`
Null Count	0	0
Mean	2024-08-09	2024-08-10 03:50
Std	—	—
Min	2022-01-03	2022-01-03 09:09
25%	2023-10-23	2023-10-23 10:09
50% (Median)	2024-08-09	2024-08-09 16:26
75%	2025-06-14	2025-06-14 09:19
Max	2026-05-11	2026-05-11 15:28

Table A.8 confirms that the operational history spans a sufficiently long period to support analyses of processing delays, shipment routing patterns, and service performance indicators.

Although this dataset was not incorporated into the forecasting models developed in this internship, it represents a potentially valuable resource for future studies focused on operational efficiency, bottleneck detection, transit time estimation, or process mining applications.

A.4.3 `delivery.parquet`

The `delivery` dataset contains information related to the final delivery stage of shipments, including delivery dates, payment transfers, beneficiary information, and monetary amounts associated with financial services.

After parsing and consolidation, the dataset comprised 241,781 observations.

Table A.9 summarizes the cardinality of the main variables available in the dataset.

Table A.9 - Cardinality summary of the `delivery` dataset

Feature	Unique values
`cab`	241431
`id`	241431
`agence_liv`	222
`date_liv`	1287
`date_transf_ccp`	1016
`montant`	6335
`beneficiaire`	14481
`n_pid`	2526
`statut`	13
`origine`	6
`etat`	3
`N_compte_Infos`	4738
`N_cheque_N_reçu_TPE`	7

The relatively large number of unique beneficiaries and financial accounts suggests that this dataset could support future studies related to customer behavior, payment mechanisms, and delivery service analysis.

Table A.10 presents descriptive statistics for the principal temporal and numerical variables.

Table A.10 - Descriptive statistics of the `delivery` dataset

Statistic	`date_liv`	`date_transf_ccp`	`montant`
Null Count	1	1	1
Mean	2024-08-21	0355-06-11	282.97
Std	—	—	1743.92
Min	2022-01-05	0001-01-01	0.0
25%	2023-10-25	0001-01-01	0.0
50% (Median)	2024-09-03	0001-01-01	0.0

Statistic	`date_liv`	`date_transf_ccp`	`montant`
75%	2025-07-03	0001-01-01	0.0
Max	2026-05-11	2026-05-09	102800.0

As shown in Table A.10, the `date_transf_ccp` variable contains placeholder dates corresponding to uninitialized values in the operational system. These observations would require additional preprocessing before being used for analytical purposes.

The highly skewed distribution of `montant` indicates that while most shipments are not associated with financial transactions, a small number involve substantial transferred amounts. Although this dataset was not incorporated into the forecasting workflow developed during the internship, it may provide valuable insights for future studies focusing on cash-on-delivery services, payment processing delays, and customer financial behavior.

A.4.4 `services.parquet`

The `services` dataset describes optional services associated with shipments, including cash collection, account-based payment services, and supplementary customer information.

After extraction and transformation, the dataset contained 205,931 observations.

Table A.11 summarizes the cardinality of the available variables.

Table A.11 - Cardinality summary of the `services` dataset

Feature	Unique values
`cab`	134293
`id`	134295
`Libelle_service`	19
`Num_compte`	4939
`Mnt_a_percevoir`	7002
`Adr/compte service`	2168
`Gsm`	20952
`Taxe_TTC`	166
`ORIGINE`	3

The diversity of service labels reflects the broad range of value-added services offered by BAM, including financial products and account-linked delivery options.

Table A.12 presents descriptive statistics for the monetary variables available in the dataset.

Table A.12 - Descriptive statistics of the `services` dataset

Statistic	`Mnt_a_percevoir`	`Taxe_TTC`
Null Count	2	3
Mean	482.14	14.80
Std	2199.64	27.09
Min	0.0	0.0

Statistic	`Mnt_a_percevoir`	`Taxe_TTC`
25%	0.0	0.0
50% (Median)	0.0	0.0
75%	300.0	35.0
Max	102800.0	999.0

The distributions presented in Table A.12 reveal a strong concentration of observations around zero values, indicating that many shipments are not associated with optional paid services. Nevertheless, the existence of high-value transactions suggests that this dataset may support future analyses related to PoD, CRBT, CEC, CCP, and other financial services offered by BAM.

Although the `delivery` and `services` datasets were not employed in the forecasting models developed during this internship, they enrich the analytical data layer and may constitute valuable resources for subsequent studies addressing customer segmentation, service utilization patterns, or financial process optimization.

Conclusion

This appendix provided a detailed overview of the data engineering activities undertaken during the internship. The acquisition process relied heavily on web automation techniques due to the absence of APIs and the operational limitations of the available information systems. Automated extraction procedures were implemented using Playwright, while HTML parsing and transformation tasks were carried out using `lxml`, Polars, and Apache Parquet.

Several datasets were investigated during the exploratory phase, including SMI exports, shipment tracking records, delivery information, and service-related data. Although not all datasets were incorporated into the final forecasting pipeline, their analysis contributed to a better understanding of the operational environment and may support future studies related to process optimization, service quality assessment, customer behavior analysis, and postal network performance evaluation.

The resulting analytical data layer constituted the foundation upon which the hierarchical forecasting system presented in Chapter 2 and Chapter 3 was developed. More generally, these activities illustrate that a substantial portion of industrial data science projects is devoted not only to model development, but also to data acquisition, validation, transformation, and long-term maintainability considerations.